

Biochemistry

“Biochemistry is the study of carbon compounds that crawl.”

—Mike Adams

“We think we have found the basic mechanism by which life comes from life.”

—Francis H. C. Crick

“The biochemistry and biophysics are the notes required for life; they conspire, collectively, to generate the real unit of life, the organism.”

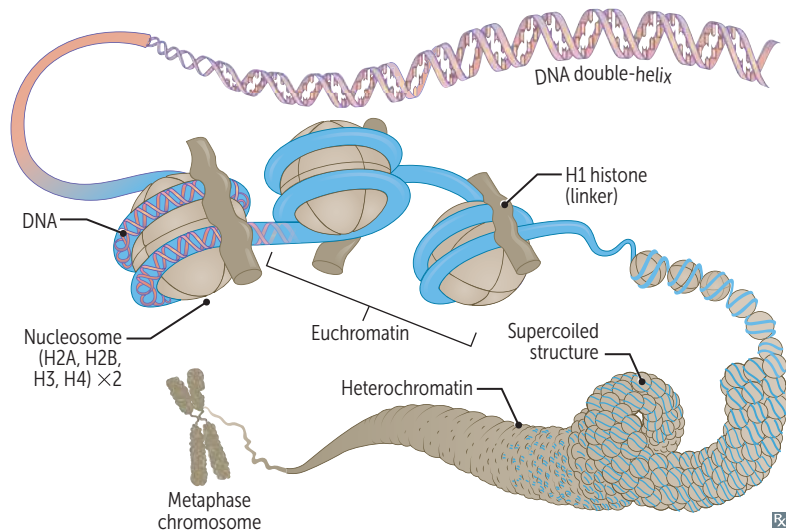
—Ursula Goodenough

This high-yield material includes molecular biology, genetics, cell biology, and principles of metabolism (especially vitamins, cofactors, minerals, and single-enzyme-deficiency diseases). When studying metabolic pathways, emphasize important regulatory steps and enzyme deficiencies that result in disease, as well as reactions targeted by pharmacologic interventions. For example, understanding the defect in Lesch-Nyhan syndrome and its clinical consequences is higher yield than memorizing every intermediate in the purine salvage pathway. Do not spend time on hard-core organic chemistry, mechanisms, or physical chemistry. Detailed chemical structures are infrequently tested; however, many structures have been included here to help students learn reactions and the important enzymes involved. Familiarity with the biochemical techniques that have medical relevance—such as ELISA, immunoelectrophoresis, Southern blotting, and PCR—is useful. Review the related biochemistry when studying pharmacology or genetic diseases as a way to reinforce and integrate the material.

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► BIOCHEMISTRY—MOLECULAR

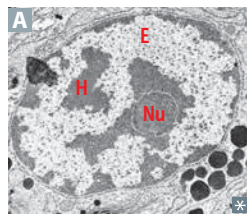
Chromatin structure



DNA exists in the condensed, chromatin form in order to fit into the nucleus. Negatively charged DNA loops twice around positively charged histone octamer to form nucleosome “**beads on a string**.” Histones are rich in the amino acids lysine and arginine. H1 binds to the nucleosome and to “linker DNA,” thereby stabilizing the chromatin fiber.

In mitosis, DNA condenses to form chromosomes. DNA and histone synthesis occur during S phase.

Heterochromatin



Condensed, appears darker on EM (labeled H in **A**). Transcriptionally inactive, sterically inaccessible. ↑ methylation, ↓ acetylation.

HeteroChromatin = **H**ighly **C**ondensed.
Barr bodies (inactive X chromosomes) are heterochromatin.

Euchromatin

Less condensed, appears lighter on EM (labeled E in **A**). Transcriptionally active, sterically accessible.

Eu = true, “truly transcribed.”
Euchromatin is **E**xpressed.

DNA methylation

Template strand cytosine and adenine are methylated in DNA replication, which allows mismatch repair enzymes to distinguish between old and new strands in prokaryotes. DNA methylation at CpG islands represses transcription.

CpG **M**ethylation **M**akes DNA **M**ute.

Histone methylation

Usually reversibly represses DNA transcription, but can activate it in some cases depending on methylation location.

Histone **M**ethylation **M**ostly **M**akes DNA **M**ute.

Histone acetylation

Relaxes DNA coiling, allowing for transcription.

Histone **A**cetylation makes DNA **A**ctive.

Nucleotides

Nucleo**S**ide = base + (deoxy)ribose (**S**ugar).

Nucleo**T**ide = base + (deoxy)ribose + phospho**T**e;
linked by 3'-5' phosphodiester bond.

PURines (**A, G**)—2 rings.

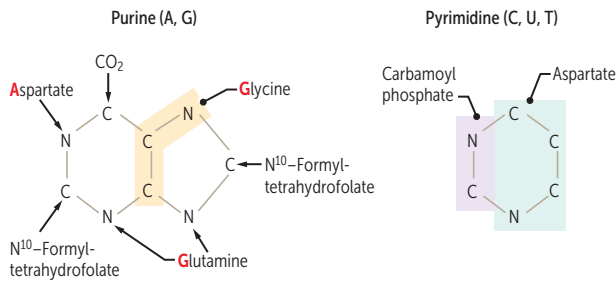
PYrimidines (**C, U, T**)—1 ring.

Deamination of cytosine makes uracil.

Deamination of adenine makes guanine.

Uracil found in RNA; thymine in DNA.

Methylation of uracil makes thymine.



5' end of incoming nucleotide bears the triphosphate (energy source for the bond).
Triphosphate bond is target of 3' hydroxyl attack.

PURe **A**s **G**old.

CUT the **PY** (pie).

Thymine has a **methy**l.

G-C bond (3 H bonds) stronger than A-T bond (2 H bonds). ↑ G-C content → ↑ melting temperature of DNA. "**C-G** bonds are like **Crazy G**lue."

GAG—Amino acids necessary for purine synthesis:

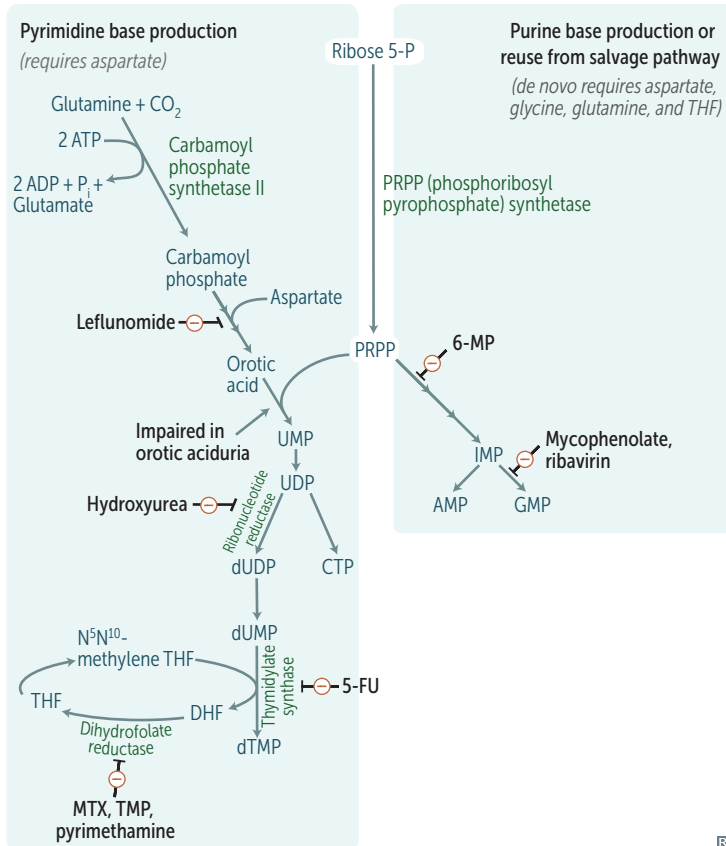
Glycine

Aspartate

Glutamine

De novo pyrimidine and purine synthesis

Various immunosuppressive, antineoplastic, and antibiotic drugs function by interfering with nucleotide synthesis:

**Disrupt pyrimidine synthesis:**

- **Leflunomide:** inhibits dihydroorotate dehydrogenase
- **Methotrexate (MTX), trimethoprim (TMP), and pyrimethamine:** inhibit dihydrofolate reductase (↓ deoxythymidine monophosphate [dTMP]) in humans, bacteria, and protozoa, respectively
- **5-fluorouracil (5-FU):** forms 5-F-dUMP, which inhibits thymidylate synthase (↓ dTMP)

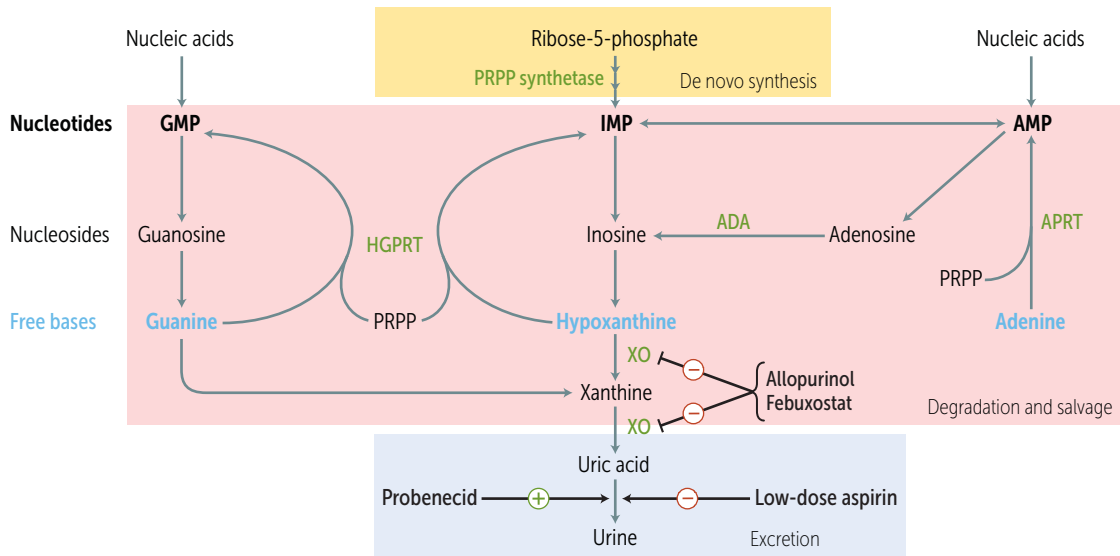
Disrupt purine synthesis:

- **6-mercaptopurine (6-MP)** and its prodrug azathioprine: inhibit de novo purine synthesis
- **Mycophenolate** and **ribavirin:** inhibit inosine monophosphate dehydrogenase

Disrupts purine and pyrimidine synthesis:

- **Hydroxyurea:** inhibits ribonucleotide reductase

Purine salvage deficiencies



ADA, adenosine deaminase; APRT, adenine phosphoribosyltransferase; HGPRT, hypoxanthine guanine phosphoribosyltransferase; XO, xanthine oxidase.

Rx

Adenosine deaminase deficiency

ADA is required for degradation of adenosine and deoxyadenosine. In ADA deficiency, $\uparrow$ dATP $\rightarrow$ toxicity in lymphocytes.

One of the major causes of autosomal recessive SCID.

Lesch-Nyhan syndrome

Defective purine salvage due to absent **HGPRT**, which converts hypoxanthine to IMP and guanine to GMP. Results in excess uric acid production and de novo purine synthesis. X-linked recessive.

Findings: intellectual disability, self-mutilation, aggression, hyperuricemia (orange “sand” [sodium urate crystals] in diaper), gout, dystonia.

Treatment: allopurinol or febuxostat (2nd line).

HGPRT:

Hyperuricemia

Gout

Pissed off (aggression, self-mutilation)

Retardation (intellectual disability)

DysTonia

Genetic code features

Unambiguous

Each codon specifies only 1 amino acid.

Degenerate/redundant

Most amino acids are coded by multiple codons.

Wobble—codons that differ in 3rd, “wobble” position may code for the same tRNA/amino acid. Specific base pairing is usually only required in the first 2 nucleotide positions of mRNA codon.

Exceptions: methionine and tryptophan encoded by only 1 codon (AUG and UGG, respectively).

Commaless, nonoverlapping

Read from a fixed starting point as a continuous sequence of bases.

Exceptions: some viruses.

Universal

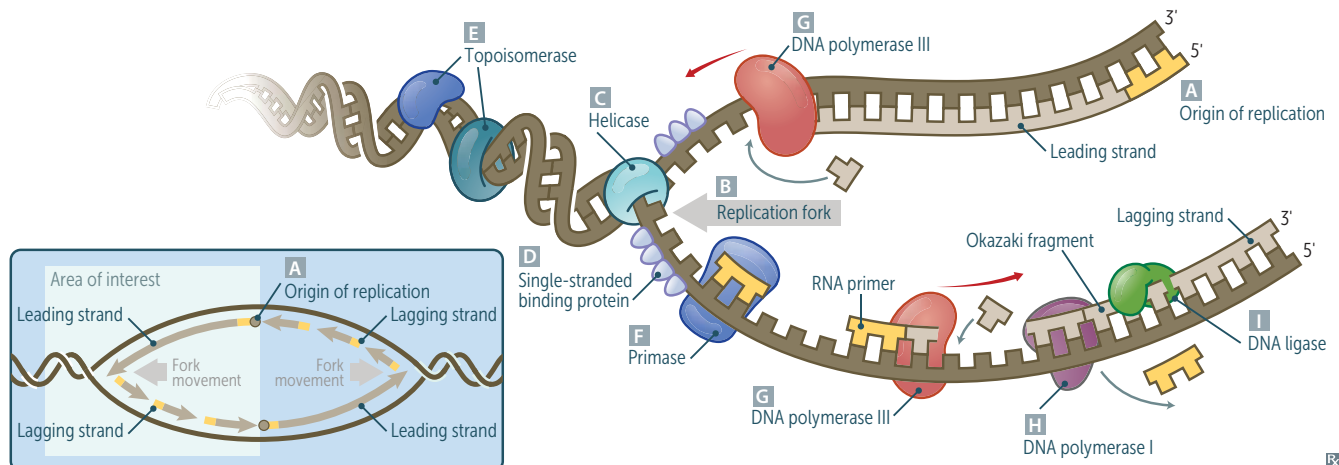
Genetic code is conserved throughout evolution.

Exception in humans: mitochondria.

DNA replication

Eukaryotic DNA replication is more complex than the prokaryotic process but uses many enzymes analogous to those listed below. In both prokaryotes and eukaryotes, DNA replication is semiconservative, involves both continuous and discontinuous (Okazaki fragment) synthesis, and occurs in the 5' → 3' direction.

Origin of replication A	Particular consensus sequence of base pairs in genome where DNA replication begins. May be single (prokaryotes) or multiple (eukaryotes).	AT-rich sequences (such as TATA box regions) are found in promoters and origins of replication.
Replication fork B	Y-shaped region along DNA template where leading and lagging strands are synthesized.	
Helicase C	Unwinds DNA template at replication fork.	
Single-stranded binding proteins D	Prevent strands from reannealing.	
DNA topoisomerases E	Create a single- or double-stranded break in the helix to add or remove supercoils.	Irinotecan/topotecan inhibit eukaryotic topoisomerase I. Etoposide/teniposide inhibit eukaryotic topoisomerase II. Fluoroquinolones inhibit prokaryotic topoisomerase II (DNA gyrase) and topoisomerase IV.
Primase F	Makes an RNA primer on which DNA polymerase III can initiate replication.	
DNA polymerase III G	Prokaryotes only. Elongates leading strand by adding deoxynucleotides to the 3' end. Elongates lagging strand until it reaches primer of preceding fragment. 3' → 5' exonuclease activity “proofreads” each added nucleotide.	DNA polymerase III has 5' → 3' synthesis and proofreads with 3' → 5' exonuclease. Drugs blocking DNA replication often have a modified 3' OH, thereby preventing addition of the next nucleotide (“chain termination”).
DNA polymerase I H	Prokaryotic only. Degrades RNA primer; replaces it with DNA.	Same functions as DNA polymerase III, also excises RNA primer with 5' → 3' exonuclease.
DNA ligase I	Catalyzes the formation of a phosphodiester bond within a strand of double-stranded DNA.	Joins Okazaki fragments.
Telomerase	Eukaryotes only. An RNA-dependent DNA polymerase that adds DNA to 3' ends of chromosomes to avoid loss of genetic material with every duplication.	Often dysregulated in cancer cells, allowing unlimited replication.



Mutations in DNA

Severity of damage: silent << missense < nonsense < frameshift.

For point (silent, missense, and nonsense) mutations:

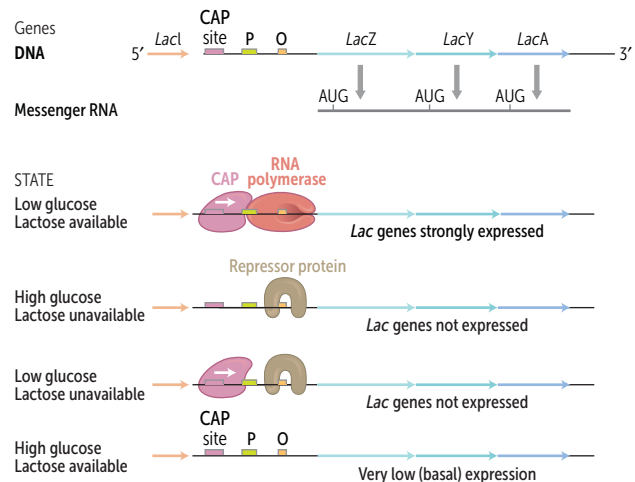
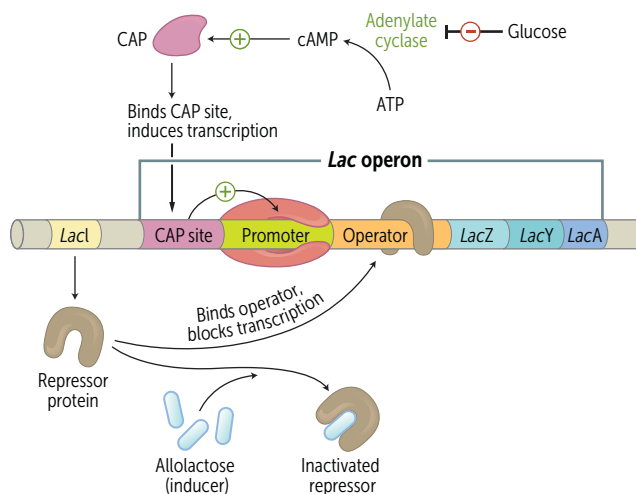
- **Transition**—purine to purine (eg, A to G) or pyrimidine to pyrimidine (eg, C to T).
- **Transversion**—purine to pyrimidine (eg, A to T) or pyrimidine to purine (eg, C to G).

Silent	Nucleotide substitution but codes for same (synonymous) amino acid; often base change in 3rd position of codon (tRNA wobble).	
Missense	Nucleotide substitution resulting in changed amino acid (called conservative if new amino acid is similar in chemical structure).	Sickle cell disease (substitution of glutamic acid with valine).
Nonsense	Nucleotide substitution resulting in early stop codon (UAG, UAA, UGA). Usually results in nonfunctional protein.	Stop the nonsense!
Frameshift	Deletion or insertion of a number of nucleotides not divisible by 3, resulting in misreading of all nucleotides downstream. Protein may be shorter or longer, and its function may be disrupted or altered.	Duchenne muscular dystrophy, Tay-Sachs disease.
Splice site	Mutation at a splice site → retained intron in the mRNA → protein with impaired or altered function.	Rare cause of cancers, dementia, epilepsy, some types of β -thalassemia.

Lac operon

Classic example of a genetic response to an environmental change. Glucose is the preferred metabolic substrate in *E. coli*, but when glucose is absent and lactose is available, the *lac* operon is activated to switch to lactose metabolism. Mechanism of shift:

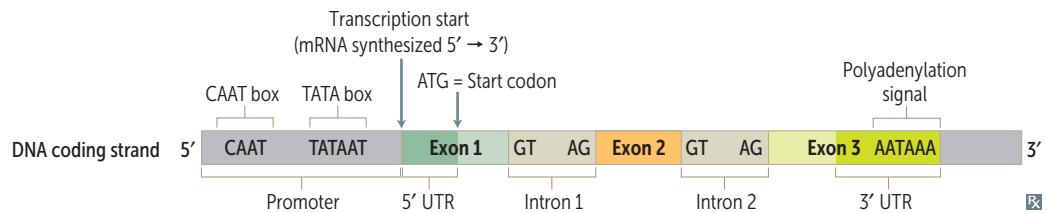
- Low glucose → ↑ adenylate cyclase activity → ↑ generation of cAMP from ATP → activation of catabolite activator protein (CAP) → ↑ transcription.
- High lactose → unbinds repressor protein from repressor/operator site → ↑ transcription.



DNA repair

Single strand		
Nucleotide excision repair	Specific endonucleases release the oligonucleotides containing damaged bases; DNA polymerase and ligase fill and reseal the gap, respectively. Repairs bulky helix-distorting lesions. Occurs in G ₁ phase of cell cycle.	Defective in xeroderma pigmentosum, which prevents repair of pyrimidine dimers that are formed as a result of ultraviolet light exposure.
Base excision repair	Base-specific G lycosylase removes altered base and creates AP site (apurinic/aprimidinic). One or more nucleotides are removed by AP- E ndonuclease, which cleaves the 5' end. L yase cleaves the 3' end. DNA P olymerase- β fills the gap and DNA L igase seals it. Occurs throughout cell cycle.	Important in repair of spontaneous/toxic deamination. “ GEL P lease”
Mismatch repair	Newly synthesized strand is recognized, mismatched nucleotides are removed, and the gap is filled and resealed. Occurs predominantly in G ₂ phase of cell cycle.	Defective in Lynch syndrome (hereditary nonpolyposis colorectal cancer [HNPCC]).
Double strand		
Nonhomologous end joining	Brings together 2 ends of DNA fragments to repair double-stranded breaks. No requirement for homology. Some DNA may be lost.	Defective in ataxia telangiectasia, breast/ovarian cancers with <i>BRCA1</i> mutation, and Fanconi anemia.
Start and stop codons		
mRNA start codons	AUG (or rarely GUG).	AUG in AUG urates protein synthesis.
Eukaryotes	Codes for methionine, which may be removed before translation is completed.	
Prokaryotes	Codes for <i>N</i> -formylmethionine (fMet).	fMet stimulates neutrophil chemotaxis.
mRNA stop codons	UGA, UAA, UAG.	UGA = U Go A way. UAA = U Are A way. UAG = U Are G one.

Functional organization of a eukaryotic gene



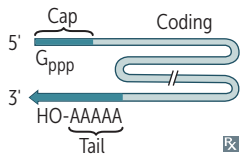
Regulation of gene expression

Promoter	Site where RNA polymerase II and multiple other transcription factors bind to DNA upstream from gene locus (AT-rich upstream sequence with TATA and CAAT boxes).	Promoter mutation commonly results in dramatic ↓ in level of gene transcription.
Enhancer	Stretch of DNA that alters gene expression by binding transcription factors (eg, activator proteins).	Enhancers and silencers may be located close to, far from, or even within (in an intron) the gene whose expression it regulates.
Silencer	Site where negative regulators (repressors) bind.	

RNA polymerases

Eukaryotes	RNA polymerase I makes rrRNA (most numerous RNA, rampant). RNA polymerase II makes mmRNA (largest RNA, massive). mRNA is read 5' to 3'. RNA polymerase III makes 5S rRNA, ttRNA (smallest RNA, tiny). No proofreading function, but can initiate chains. RNA polymerase II opens DNA at promoter site.	I, II, and III are numbered in the same order that their products are used in protein synthesis: rRNA, mRNA, then tRNA. α-amanitin, found in <i>Amanita phalloides</i> (death cap mushrooms), inhibits RNA polymerase II. Causes severe hepatotoxicity if ingested. Actinomycin D inhibits RNA polymerase in both prokaryotes and eukaryotes.
Prokaryotes	1 RNA polymerase (multisubunit complex) makes all 3 kinds of RNA.	Rifampin inhibits DNA-dependent RNA polymerase in prokaryotes.

RNA processing (eukaryotes)



Initial transcript is called heterogeneous nuclear RNA (hnRNA). hnRNA is then modified and becomes mRNA.

The following processes occur in the nucleus:

- Capping of 5' end (addition of 7-methylguanosine cap)
- Polyadenylation of 3' end (≈ 200 A's)
- Splicing out of introns

Capped, tailed, and spliced transcript is called mRNA.

mRNA is transported out of the nucleus into the cytosol, where it is translated.

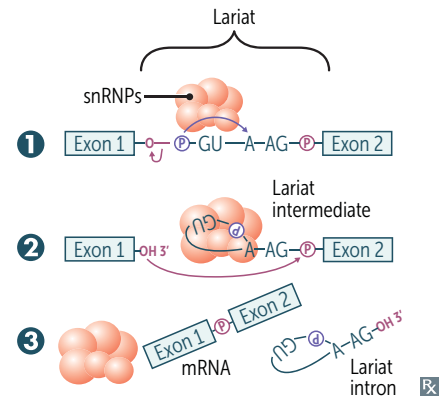
mRNA quality control occurs at cytoplasmic processing bodies (P-bodies), which contain exonucleases, decapping enzymes, and microRNAs; mRNAs may be stored in P-bodies for future translation.

Poly-A polymerase does not require a template. AAUAAA = polyadenylation signal.

Splicing of pre-mRNA

- 1 Primary transcript combines with small nuclear ribonucleoproteins (snRNPs) and other proteins to form spliceosome.
- 2 Lariat-shaped (looped) intermediate is generated.
- 3 Lariat is released to precisely remove intron and join 2 exons.

Antibodies to spliceosomal snRNPs (anti-Smith antibodies) are highly specific for SLE. Anti-U1 RNP antibodies are highly associated with mixed connective tissue disease (MCTD).

**Introns vs exons**

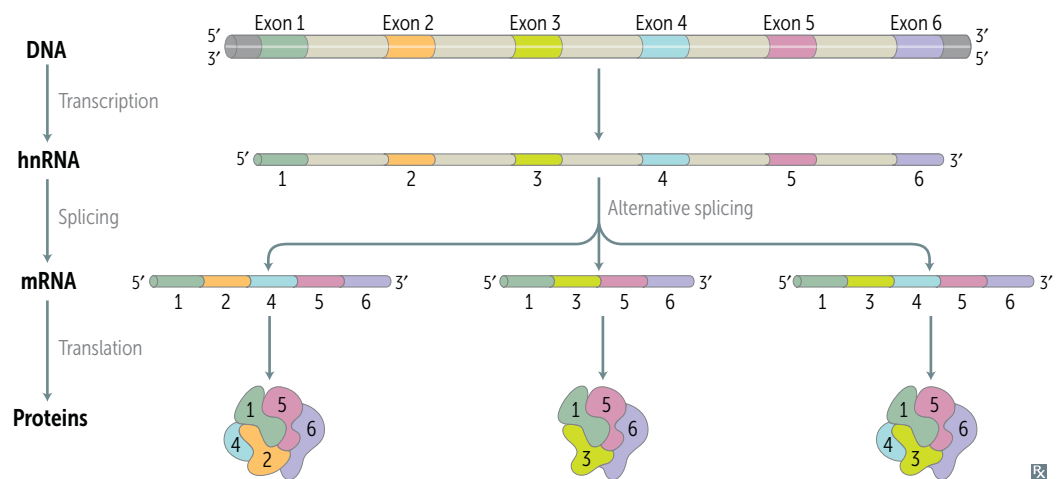
Exons contain the actual genetic information coding for protein.

Introns are intervening noncoding segments of DNA.

Different exons are frequently combined by alternative splicing to produce a larger number of unique proteins.

Introns are **in**tervening sequences and stay **in** the nucleus, whereas **ex**ons **ex**it and are **ex**pressed.

Abnormal splicing variants are implicated in oncogenesis and many genetic disorders (eg, β -thalassemia).

**microRNAs**

MicroRNAs (miRNA) are small, conserved, noncoding RNA molecules that posttranscriptionally regulate gene expression by targeting the 3' untranslated region of specific mRNAs for degradation or translational repression. Abnormal expression of miRNAs contributes to certain malignancies (eg, by silencing an mRNA from a tumor suppressor gene).

tRNA**Structure**

75–90 nucleotides, 2° structure, cloverleaf form, anticodon end is opposite 3' aminoacyl end. All tRNAs, both eukaryotic and prokaryotic, have CCA at 3' end along with a high percentage of chemically modified bases. The amino acid is covalently bound to the 3' end of the tRNA. **CCA Can Carry Amino acids.**

T-arm: contains the TΨC (ribothymidine, pseudouridine, cytidine) sequence necessary for tRNA-ribosome binding. **T-arm Tethers** tRNA molecule to ribosome.

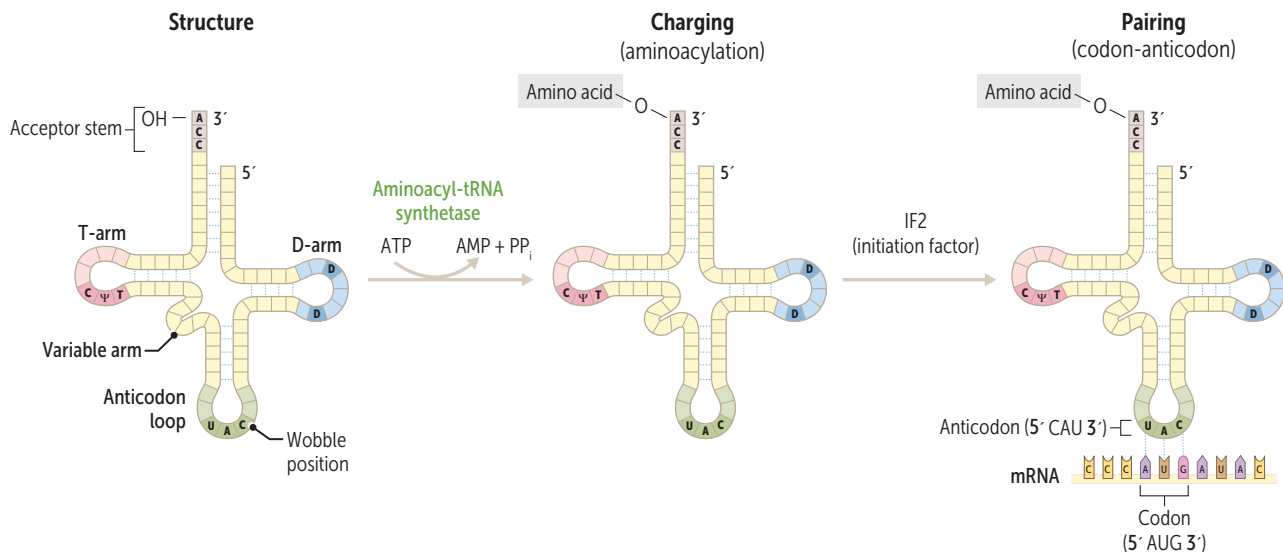
D-arm: contains dihydrouridine residues necessary for tRNA recognition by the correct aminoacyl-tRNA synthetase. **D-arm Detects** the tRNA by aminoacyl-tRNA synthetase.

Acceptor stem: the 5'-CCA-3' is the amino acid acceptor site.

Charging

Aminoacyl-tRNA synthetase (1 per amino acid; “matchmaker”; uses ATP) scrutinizes amino acid before and after it binds to tRNA. If incorrect, bond is hydrolyzed. The amino acid-tRNA bond has energy for formation of peptide bond. A mischarged tRNA reads usual codon but inserts wrong amino acid.

Aminoacyl-tRNA synthetase and binding of charged tRNA to the codon are responsible for accuracy of amino acid selection.



Protein synthesis**Initiation**

Initiated by GTP hydrolysis; initiation factors (eukaryotic IFs) help assemble the 40S ribosomal subunit with the initiator tRNA and are released when the mRNA and the ribosomal 60S subunit assemble with the complex.

Eukaryotes: 40S + 60S → 80S (**E**ven).
PrOkaryotes: 30S + 50S → 70S (**O**dd).
 Synthesis occurs from N-terminus to C-terminus.

ATP—tRNA **A**ctivation (charging).

GTP—tRNA **G**ripping and **G**oing places (translocation).

Elongation

1. Aminoacyl-tRNA binds to A site (except for initiator methionine)
2. rRNA (“ribozyme”) catalyzes peptide bond formation, transfers growing polypeptide to amino acid in A site
3. Ribosome advances 3 nucleotides toward 3' end of mRNA, moving peptidyl tRNA to P site (translocation)

Think of “going **APE**”:

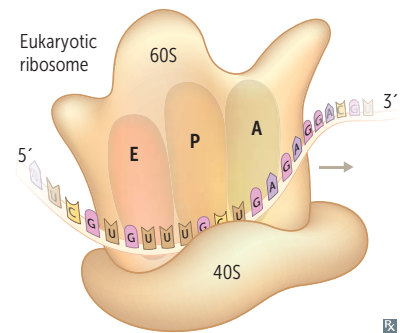
A site = incoming **A**minoacyl-tRNA.

P site = accommodates growing **P**eptide.

E site = holds **E**mpy tRNA as it **E**xits.

Termination

Stop codon is recognized by release factor, and completed polypeptide is released from ribosome.

**Posttranslational modifications****Trimming**

Removal of N- or C-terminal propeptides from zymogen to generate mature protein (eg, trypsinogen to trypsin).

Covalent alterations

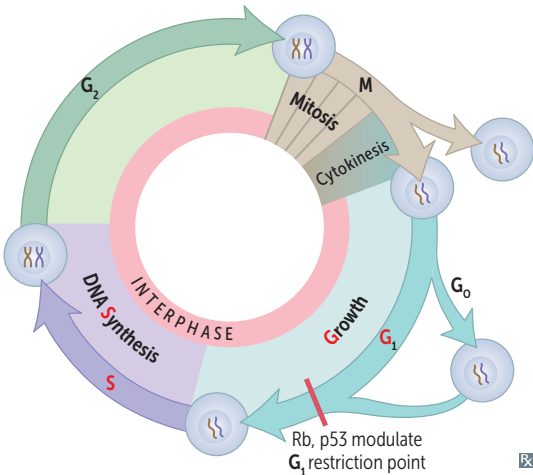
Phosphorylation, glycosylation, hydroxylation, methylation, acetylation, and ubiquitination.

Chaperone protein

Intracellular protein involved in facilitating and/or maintaining protein folding. For example, in yeast, heat shock proteins (eg, HSP60) are expressed at high temperatures to prevent protein denaturing/misfolding.

▶ BIOCHEMISTRY—CELLULAR

Cell cycle phases Checkpoints control transitions between phases of cell cycle. This process is regulated by cyclins, cyclin-dependent kinases (CDKs), and tumor suppressors. M phase (shortest phase of cell cycle) includes mitosis (prophase, prometaphase, metaphase, anaphase, telophase) and cytokinesis (cytoplasm splits in two). G_1 and G_0 are of variable duration.

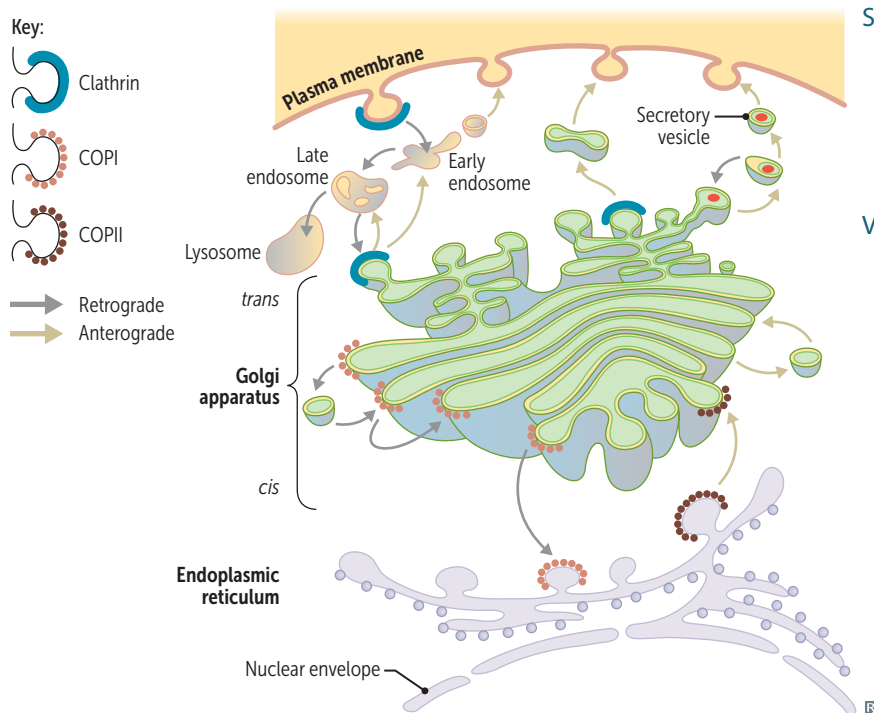
REGULATION OF CELL CYCLE		
Cyclin-dependent kinases	Constitutive and inactive.	
Cyclins	Regulatory proteins that control cell cycle events; phase specific; activate CDKs.	
Cyclin-CDK complexes	Phosphorylate other proteins to coordinate cell cycle progression; must be activated and inactivated at appropriate times for cell cycle to progress.	
Tumor suppressors	p53 induces p21, which inhibits CDKs → hypophosphorylation (activation) of Rb → inhibition of G ₁ -S progression. Mutations in tumor suppressor genes can result in unrestrained cell division (eg, Li-Fraumeni syndrome).	
		
CELL TYPES		
Permanent	Remain in G ₀ , regenerate from stem cells.	Neurons, skeletal and cardiac muscle, RBCs.
Stable (quiescent)	Enter G ₁ from G ₀ when stimulated.	Hepatocytes, lymphocytes.
Labile	Never go to G ₀ , divide rapidly with a short G ₁ . Most affected by chemotherapy.	Bone marrow, gut epithelium, skin, hair follicles, germ cells.
Rough endoplasmic reticulum	Site of synthesis of secretory (exported) proteins and of N-linked oligosaccharide addition to many proteins. Nissl bodies (RER in neurons)—synthesize peptide neurotransmitters for secretion. Free ribosomes—unattached to any membrane; site of synthesis of cytosolic and organellar proteins.	Mucus-secreting goblet cells of the small intestine and antibody-secreting plasma cells are rich in RER.
Smooth endoplasmic reticulum	Site of steroid synthesis and detoxification of drugs and poisons. Lacks surface ribosomes.	Liver hepatocytes and steroid hormone-producing cells of the adrenal cortex and gonads are rich in SER.

Cell trafficking

Golgi is the distribution center for proteins and lipids from the ER to the vesicles and plasma membrane. Modifies N-oligosaccharides on asparagine. Adds O-oligosaccharides on serine and threonine. Adds mannose-6-phosphate to proteins for trafficking to lysosomes.

Endosomes are sorting centers for material from outside the cell or from the Golgi, sending it to lysosomes for destruction or back to the membrane/Golgi for further use.

I-cell disease (inclusion cell disease/mucopolidosis type II)—inherited lysosomal storage disorder; defect in *N*-acetylglucosaminyl-1-phosphotransferase → failure of the Golgi to phosphorylate mannose residues (ie, ↓ mannose-6-phosphate) on glycoproteins → proteins are secreted extracellularly rather than delivered to lysosomes. Results in coarse facial features, clouded corneas, restricted joint movement, and high plasma levels of lysosomal enzymes. Often fatal in childhood.



Signal recognition particle (SRP)

Abundant, cytosolic ribonucleoprotein that traffics proteins from the ribosome to the RER. Absent or dysfunctional SRP → proteins accumulate in the cytosol.

Vesicular trafficking proteins

COPI: Golgi → Golgi (retrograde); *cis*-Golgi → ER.

COPII: ER → *cis*-Golgi (anterograde).

“**Two** (COPII) steps forward (anterograde); **one** (COPI) step back (retrograde).”

Clathrin: *trans*-Golgi → lysosomes; plasma membrane → endosomes (receptor-mediated endocytosis [eg, LDL receptor activity]).

Peroxisome

Membrane-enclosed organelle involved in catabolism of very-long-chain fatty acids (through β -oxidation), branched-chain fatty acids, amino acids, and ethanol.

Peroxisomal disorders commonly lead to neurologic diseases due to deficits in synthesis of plasmalogens, important phospholipids in myelin. Peroxisomal diseases include Zellweger syndrome (hypotonia, seizures, hepatomegaly, early death) and Refsum disease (scaly skin, ataxia, cataracts/night blindness, shortening of 4th toe, epiphyseal dysplasia).

Proteasome

Barrel-shaped protein complex that degrades damaged or ubiquitin-tagged proteins. Defects in the ubiquitin-proteasome system have been implicated in some cases of Parkinson disease.

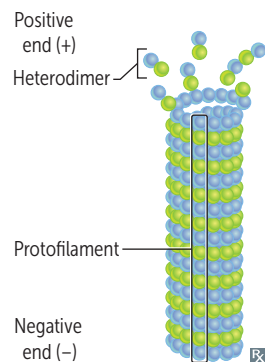
Cytoskeletal elements A network of protein fibers within the cytoplasm that supports cell structure, cell and organelle movement, and cell division.

TYPE OF FILAMENT	PREDOMINANT FUNCTION	EXAMPLES
Microfilaments	Muscle contraction, cytokinesis	Actin, microvilli.
Intermediate filaments	Maintain cell structure	Vimentin, desmin, cytokeratin, lamins, glial fibrillary acid proteins (GFAP), neurofilaments.
Microtubules	Movement, cell division	Cilia, flagella, mitotic spindle, axonal trafficking, centrioles.

Immunohistochemical stains for intermediate filaments

STAIN	CELL TYPE	IDENTIFIES
Vimentin	M esenchymal tissue (eg, fibroblasts, endothelial cells, macrophages)	M esenchymal tumors (eg, sarcoma), but also many other tumors (eg, endometrial carcinoma, renal cell carcinoma, meningioma)
DesMin	M uscle	Muscle tumors (eg, rhabdomyosarcoma)
Cytokeratin	Epithelial cells	Epithelial tumors (eg, squamous cell carcinoma)
GFAP	Neuro G lia (eg, astrocytes, Schwann cells, oligodendrocytes)	Astrocytoma, G lioblastoma
Neurofilaments	Neurons	Neuronal tumors (eg, neuroblastoma)

Microtubule



Cylindrical outer structure composed of a helical array of polymerized heterodimers of α - and β -tubulin. Each dimer has 2 GTP bound. Incorporated into flagella, cilia, mitotic spindles. Grows slowly, collapses quickly. Also involved in slow axoplasmic transport in neurons.

Molecular motor proteins—transport cellular cargo toward opposite ends of microtubule tracks.

- Dynein—retrograde to microtubule (+ $\rightarrow$ -).
- Kinesin—anterograde to microtubule (- $\rightarrow$ +).

Drugs that act on microtubules (**M**icrotubules **G**et **C**onstructed **V**ery **P**oorly):

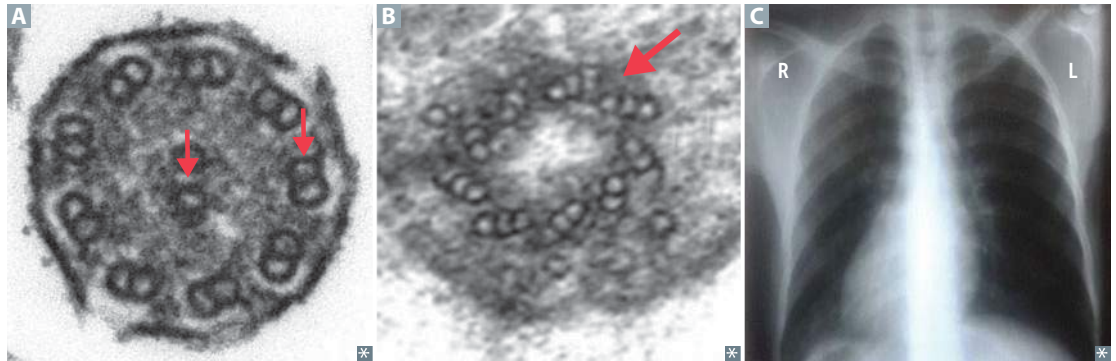
- **M**ebendazole (antihelminthic)
- **G**riseofulvin (antifungal)
- **C**olchicine (antigout)
- **V**incristine/**V**inblastine (anticancer)
- **P**aclitaxel (anticancer)

Negative end **N**ear **N**ucleus
Positive end **P**oints to **P**eriphery

Cilia structure

9 doublet + 2 singlet arrangement of microtubules (arrows in **A**).
 Basal body (base of cilium below cell membrane) consists of 9 microtubule triplets (arrow in **B**) with no central microtubules.
 Axonemal dynein—ATPase that links peripheral 9 doublets and causes bending of cilium by differential sliding of doublets.

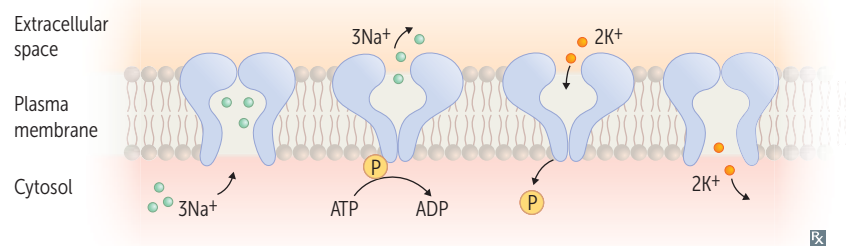
Kartagener syndrome (1° ciliary dyskinesia)—immotile cilia due to a dynein arm defect. Results in ↓ male and female fertility due to immotile sperm and dysfunctional fallopian tube cilia, respectively; ↑ risk of ectopic pregnancy. Can cause bronchiectasis, recurrent sinusitis, chronic ear infections, conductive hearing loss, and situs inversus (eg, dextrocardia on CXR **C**).

**Sodium-potassium pump**

Na^+ - K^+ ATPase is located in the plasma membrane with ATP site on cytosolic side. For each ATP consumed, 3Na^+ go out of the cell (pump phosphorylated) and 2K^+ come into the cell (pump dephosphorylated). Plasma membrane is an asymmetric lipid bilayer containing cholesterol, phospholipids, sphingolipids, glycolipids, and proteins.

Pumpkin = pump K^+ in.

Ouabain inhibits by binding to K^+ site. Cardiac glycosides (digoxin and digitoxin) directly inhibit the Na^+ - K^+ ATPase, which leads to indirect inhibition of Na^+ / Ca^{2+} exchange → ↑ $[\text{Ca}^{2+}]_i$ → ↑ cardiac contractility.

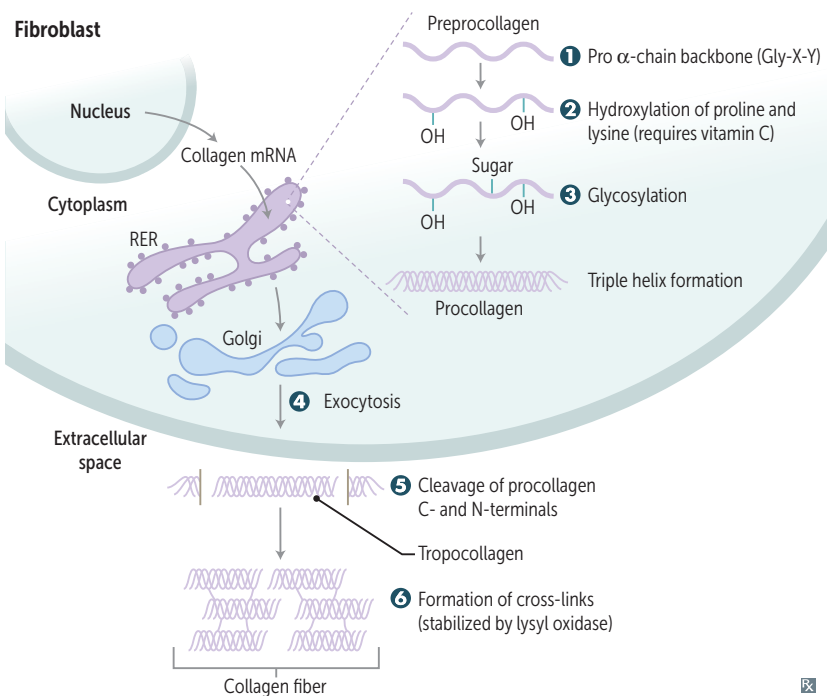


Collagen

Most abundant protein in the human body.
Extensively modified by posttranslational modification.
Organizes and strengthens extracellular matrix.

Be (So Totally) Cool, Read Books.

Type I	Most common (90%)— B one (made by osteoblasts), S kin, T endon, dentin, fascia, cornea, late wound repair.	Type I : b one. ↓ production in osteogenesis imperfecta type I.
Type II	C artilage (including hyaline), vitreous body, nucleus pulposus.	Type II : cartwo l age.
Type III	R eticulin—skin, b lood v essels, uterus, fetal tissue, granulation tissue.	Type III : deficient in the uncommon, v ascular type of E hlers- D anlos syndrome (ThreE D).
Type IV	B asement membrane, basal lamina, lens.	Type IV : under the f loor (basement membrane). Defective in Alport syndrome; targeted by autoantibodies in Goodpasture syndrome.

Collagen synthesis and structure

1 Synthesis—translation of collagen α chains (preprocollagen)—usually Gly-X-Y (X and Y are proline or lysine). Glycine content best reflects collagen synthesis (collagen is $\frac{1}{3}$ glycine).

2 Hydroxylation—hydroxylation of specific proline and lysine residues. Requires vitamin C; deficiency $\rightarrow$ scurvy.

3 Glycosylation—glycosylation of pro- α -chain hydroxylysine residues and formation of procollagen via hydrogen and disulfide bonds (triple helix of 3 collagen α chains). Problems forming triple helix $\rightarrow$ osteogenesis imperfecta.

4 Exocytosis—exocytosis of procollagen into extracellular space.

5 Proteolytic processing—cleavage of disulfide-rich terminal regions of procollagen $\rightarrow$ insoluble tropocollagen. Problems with cleavage $\rightarrow$ Ehlers-Danlos syndrome.

6 Cross-linking—reinforcement of many staggered tropocollagen molecules by covalent lysine-hydroxylysine cross-linkage (by copper-containing lysyl oxidase) to make collagen fibrils. Problems with cross-linking $\rightarrow$ Ehlers-Danlos syndrome, Menkes disease.

Osteogenesis imperfecta

Genetic bone disorder (brittle bone disease) caused by a variety of gene defects (most commonly *COL1A1* and *COL1A2*).

Most common form is autosomal dominant with ↓ production of otherwise normal type I collagen. Manifestations can include:

- Multiple fractures with minimal trauma **A B**; may occur during the birth process
- Blue sclerae **C** due to the translucent connective tissue over choroidal veins
- Some forms have tooth abnormalities, including opalescent teeth that wear easily due to lack of dentin (dentinogenesis imperfecta)
- Hearing loss (abnormal ossicles)

May be confused with child abuse.

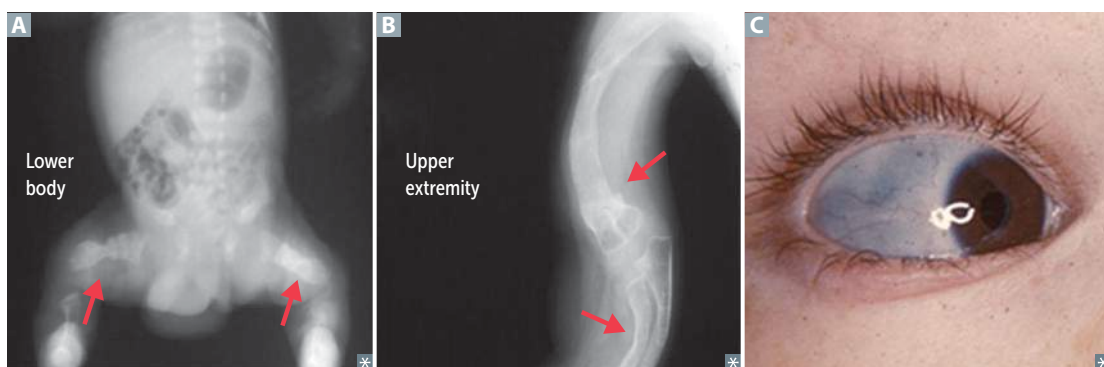
Patients can't **BITE**:

Bones = multiple fractures

I (eye) = blue sclerae

Teeth = dental imperfections

Ear = hearing loss



Ehlers-Danlos syndrome



Faulty collagen synthesis causing hyperextensible skin, tendency to bleed (easy bruising), and hypermobile joints **A**.

Multiple types. Inheritance and severity vary. Can be autosomal dominant or recessive. May be associated with joint dislocation, berry and aortic aneurysms, organ rupture.

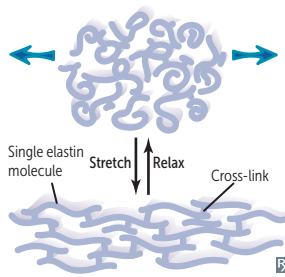
Hypermobility type (joint instability): most common type.

Classical type (joint and skin symptoms): caused by a mutation in type V collagen (eg, *COL5A1*, *COL5A2*).

Vascular type (vascular and organ rupture): deficient type III collagen.

Menkes disease

X-linked recessive connective tissue disease caused by impaired copper absorption and transport due to defective Menkes protein (*ATP7A*). Leads to ↓ activity of lysyl oxidase (copper is a necessary cofactor). Results in brittle, “kinky” hair, growth retardation, and hypotonia.

Elastin

Stretchy protein within skin, lungs, large arteries, elastic ligaments, vocal cords, ligamenta flava (connect vertebrae → relaxed and stretched conformations).

Rich in nonhydroxylated proline, glycine, and lysine residues.

Tropoelastin with fibrillin scaffolding.

Cross-linking takes place extracellularly and gives elastin its elastic properties.

Broken down by elastase, which is normally inhibited by α_1 -antitrypsin.

α_1 -Antitrypsin deficiency results in excess elastase activity, which can cause emphysema.

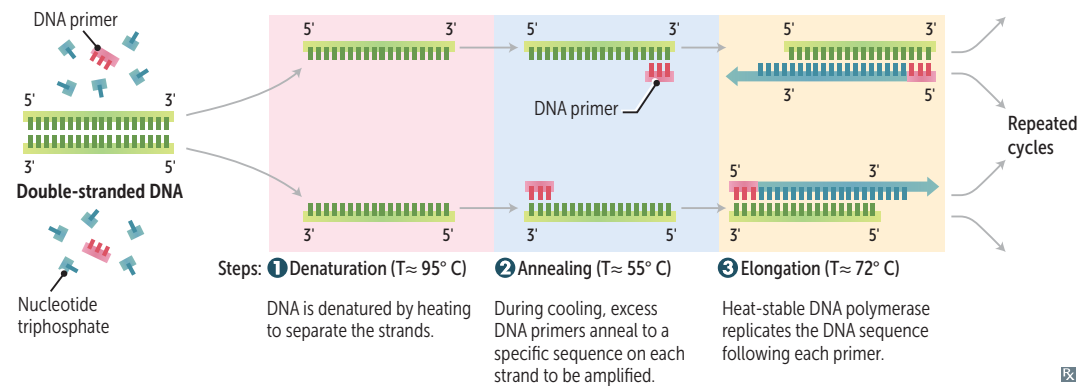
Wrinkles of aging are due to ↓ collagen and elastin production.

Marfan syndrome—autosomal dominant connective tissue disorder affecting skeleton, heart, and eyes. *FBNI* gene mutation on chromosome 15 results in defective fibrillin, a glycoprotein that forms a sheath around elastin. Findings: tall with long extremities; pectus carinatum (more specific) or pectus excavatum; hypermobile joints; long, tapering fingers and toes (arachnodactyly); cystic medial necrosis of aorta; aortic incompetence and dissecting aortic aneurysms; floppy mitral valve. Subluxation of lenses, typically upward and temporally.

► BIOCHEMISTRY—LABORATORY TECHNIQUES

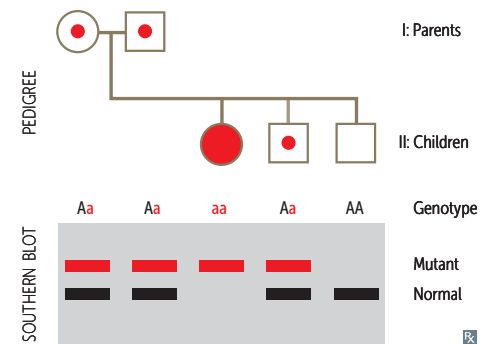
Polymerase chain reaction

Molecular biology laboratory procedure used to amplify a desired fragment of DNA. Useful as a diagnostic tool (eg, neonatal HIV, herpes encephalitis).



Blotting procedures**Southern blot**

1. DNA sample is enzymatically cleaved into smaller pieces, which are separated on a gel by electrophoresis, and then transferred to a filter.
2. Filter is exposed to radiolabeled DNA probe that recognizes and anneals to its complementary strand.
3. Resulting double-stranded, labeled piece of DNA is visualized when filter is exposed to film.

**Northern blot**

Similar to Southern blot, except that an **RNA** sample is electrophoresed. Useful for studying mRNA levels, which are reflective of gene expression.

SNOW DRoP:

Southern = **D**NA

Northern = **R**NA

Western = **P**rotein

Western blot

Sample protein is separated via gel electrophoresis and transferred to a membrane. Labeled antibody is used to bind to relevant **protein** (eg, confirmatory test for HIV after ⊕ ELISA).

Southwestern blot

Identifies **DNA-binding proteins** (eg, transcription factors) using labeled oligonucleotide probes.

Flow cytometry

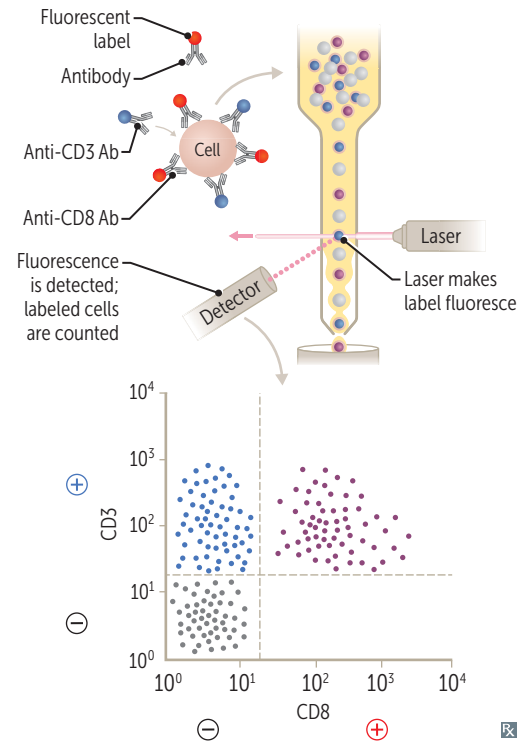
Laboratory technique to assess size, granularity, and protein expression (immunophenotype) of individual cells in a sample.

Cells are tagged with antibodies specific to surface or intracellular proteins. Antibodies are then tagged with a unique fluorescent dye. Sample is analyzed one cell at a time by focusing a laser on the cell and measuring light scatter and intensity of fluorescence.

Data are plotted either as histogram (one measure) or scatter plot (any two measures, as shown). In illustration:

- Cells in left lower quadrant $\ominus$ for both CD8 and CD3.
- Cells in right lower quadrant $\oplus$ for CD8 and $\ominus$ for CD3. Right lower quadrant is empty because all CD8-expressing cells also express CD3.
- Cells in left upper quadrant $\oplus$ for CD3 and $\ominus$ for CD8.
- Cells in right upper quadrant $\oplus$ for CD8 and CD3 (red + blue $\rightarrow$ purple).

Commonly used in workup of hematologic abnormalities (eg, paroxysmal nocturnal hemoglobinuria, fetal RBCs in mother's blood) and immunodeficiencies (eg, CD4 cell count in HIV).



Microarrays

Thousands of nucleic acid sequences are arranged in grids on glass or silicon. DNA or RNA probes are hybridized to the chip, and a scanner detects the relative amounts of complementary binding. Used to profile gene expression levels of thousands of genes simultaneously to study certain diseases and treatments. Able to detect single nucleotide polymorphisms (SNPs) and copy number variations (CNVs) for a variety of applications including genotyping, clinical genetic testing, forensic analysis, cancer mutations, and genetic linkage analysis.

Enzyme-linked immunosorbent assay

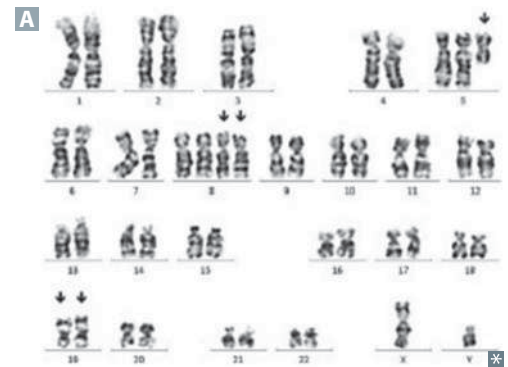
Immunologic test used to detect the presence of either a specific antigen (eg, HBsAg) or antibody (eg, anti-HBs) in a patient's blood sample. Detection involves the use of an antibody linked to an enzyme. Added substrate reacts with enzyme, producing a detectable signal. Can have high sensitivity and specificity, but is less specific than Western blot.

Karyotyping

A process in which metaphase chromosomes are stained, ordered, and numbered according to morphology, size, arm-length ratio, and banding pattern (arrows in **A** point to extensive abnormalities in a cancer cell).

Can be performed on a sample of blood, bone marrow, amniotic fluid, or placental tissue.

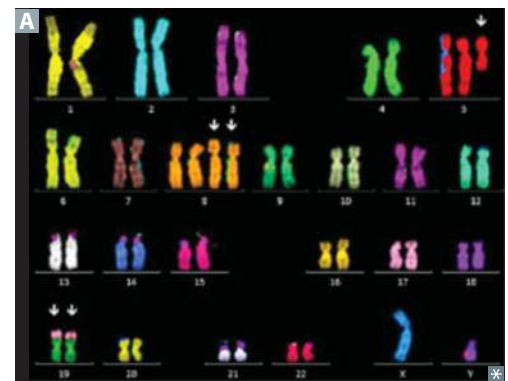
Used to diagnose chromosomal imbalances (eg, autosomal trisomies, sex chromosome disorders).

**Fluorescence in situ hybridization**

Fluorescent DNA or RNA probe binds to specific gene site of interest on chromosomes (arrows in **A** point to abnormalities in a cancer cell, whose karyotype is seen above; each fluorescent color represents a chromosome-specific probe).

Used for specific localization of genes and direct visualization of chromosomal anomalies at the molecular level.

- Microdeletion—no fluorescence on a chromosome compared to fluorescence at the same locus on the second copy of that chromosome
- Translocation—fluorescence signal that corresponds to one chromosome is found in a different chromosome
- Duplication—extra site of fluorescence on one chromosome relative to its homologous chromosome

**Cloning methods**

Cloning is the production of a recombinant DNA molecule that is self-perpetuating.

Steps:

1. Isolate eukaryotic mRNA (post-RNA processing) of interest.
2. Expose mRNA to reverse transcriptase to produce cDNA (lacks introns).
3. Insert cDNA fragments into bacterial plasmids containing antibiotic resistance genes.
4. Transform recombinant plasmid into bacteria.
5. Surviving bacteria on antibiotic medium produce cloned DNA (copies of cDNA).

Gene expression modifications

Transgenic strategies in mice involve:

- Random insertion of gene into mouse genome
- Targeted insertion or deletion of gene through homologous recombination with mouse gene

Knock-**out** = removing a gene, taking it **out**.
Knock-**in** = **in**serting a gene.

Cre-lox system

Can inducibly manipulate genes at specific developmental points (eg, to study a gene whose deletion causes embryonic death).

RNA interference

dsRNA is synthesized that is complementary to the mRNA sequence of interest. When transfected into human cells, dsRNA separates and promotes degradation of target mRNA, “knocking down” gene expression.

► BIOCHEMISTRY—GENETICS

Genetic terms

TERM	DEFINITION	EXAMPLE
Codominance	Both alleles contribute to the phenotype of the heterozygote.	Blood groups A, B, AB; α_1 -antitrypsin deficiency.
Variable expressivity	Patients with the same genotype have varying phenotypes.	2 patients with neurofibromatosis type 1 (NF1) may have varying disease severity.
Incomplete penetrance	Not all individuals with a mutant genotype show the mutant phenotype.	<i>BRCA1</i> gene mutations do not always result in breast or ovarian cancer.
Pleiotropy	One gene contributes to multiple phenotypic effects.	Untreated phenylketonuria (PKU) manifests with light skin, intellectual disability, and musty body odor.
Anticipation	Increased severity or earlier onset of disease in succeeding generations.	Trinucleotide repeat diseases (eg, Huntington disease).
Loss of heterozygosity	If a patient inherits or develops a mutation in a tumor suppressor gene, the complementary allele must be deleted/mutated before cancer develops. This is not true of oncogenes.	Retinoblastoma and the “two-hit hypothesis,” Lynch syndrome (HNPCC), Li-Fraumeni syndrome.
Dominant negative mutation	Exerts a dominant effect. A heterozygote produces a nonfunctional altered protein that also prevents the normal gene product from functioning.	Mutation of a transcription factor in its allosteric site. Nonfunctioning mutant can still bind DNA, preventing wild-type transcription factor from binding.
Linkage disequilibrium	Tendency for certain alleles at 2 linked loci to occur together more or less often than expected by chance. Measured in a population, not in a family, and often varies in different populations.	

Genetic terms (continued)

TERM	DEFINITION	EXAMPLE
Mosaicism	Presence of genetically distinct cell lines in the same individual. Somatic mosaicism—mutation arises from mitotic errors after fertilization and propagates through multiple tissues or organs. Gonadal mosaicism—mutation only in egg or sperm cells. If parents and relatives do not have the disease, suspect gonadal (or germline) mosaicism.	McCune-Albright syndrome —due to mutation affecting G-protein signaling. Presents with unilateral café-au-lait spots with ragged edges, polyostotic fibrous dysplasia, and at least one endocrinopathy (eg, precocious puberty). Lethal if mutation occurs before fertilization (affecting all cells), but survivable in patients with mosaicism.
Locus heterogeneity	Mutations at different loci can produce a similar phenotype.	Albinism.
Allelic heterogeneity	Different mutations in the same locus produce the same phenotype.	β-thalassemia.
Heteroplasmy	Presence of both normal and mutated mtDNA, resulting in variable expression in mitochondrially inherited disease.	
Uniparental disomy	Offspring receives 2 copies of a chromosome from 1 parent and no copies from the other parent. Heterodisomy (heterozygous) indicates a meiosis I error. Isodisomy (homozygous) indicates a meiosis II error or postzygotic chromosomal duplication of one of a pair of chromosomes, and loss of the other of the original pair.	Uniparental is euploid (correct number of chromosomes), not aneuploid. Most occurrences of uniparental disomy (UPD) → normal phenotype. Consider UPD in an individual manifesting a recessive disorder when only one parent is a carrier.

Hardy-Weinberg population genetics

	pA	qa
pA	AA $p \times p = p^2$	Aa $p \times q$
qa	Aa $p \times q$	aa $q \times q = q^2$

If a population is in Hardy-Weinberg equilibrium and if p and q are the frequencies of separate alleles, then: $p^2 + 2pq + q^2 = 1$ and $p + q = 1$, which implies that:
 p^2 = frequency of homozygosity for allele A
 q^2 = frequency of homozygosity for allele a
 $2pq$ = frequency of heterozygosity (carrier frequency, if an autosomal recessive disease).
 The frequency of an X-linked recessive disease in males = q and in females = q^2 .

Hardy-Weinberg law assumptions include:

- No mutation occurring at the locus
- Natural selection is not occurring
- Completely random mating
- No net migration

Imprinting

At some loci, only one allele is active; the other is inactive (imprinted/inactivated by methylation). With one allele inactivated, deletion of the active allele → disease.

Both Prader-Willi and Angelman syndromes are due to mutation or deletion of genes on chromosome 15.

Prader-Willi syndrome

Maternal imprinting: gene from mom is normally silent and **P**aternal gene is deleted/mutated. Results in hyperphagia, obesity, intellectual disability, hypogonadism, and hypotonia.

25% of cases due to maternal uniparental disomy (two maternally imprinted genes are received; no paternal gene received).

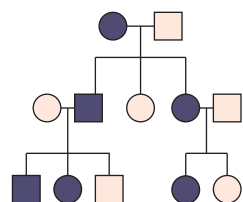
AngelMan** syndrome**

Paternal imprinting: gene from dad is normally silent and **M**aternal gene is deleted/mutated. Results in inappropriate laughter (“happy puppet”), seizures, ataxia, and severe intellectual disability.

5% of cases due to paternal uniparental disomy (two paternally imprinted genes are received; no maternal gene received).

Modes of inheritance

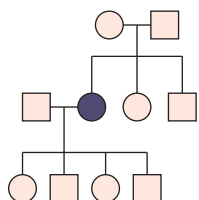
Autosomal dominant



Often due to defects in structural genes. Many generations, both males and females are affected.

Often pleiotropic (multiple apparently unrelated effects) and variably expressive (different between individuals). Family history crucial to diagnosis. With one affected (heterozygous) parent, on average, $\frac{1}{2}$ of children affected.

Autosomal recessive



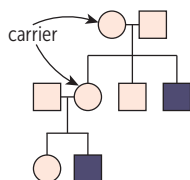
Often due to enzyme deficiencies. Usually seen in only 1 generation.

Commonly more severe than dominant disorders; patients often present in childhood.

↑ risk in consanguineous families.

With 2 carrier (heterozygous) parents, on average: $\frac{1}{4}$ of children will be affected (homozygous), $\frac{1}{2}$ of children will be carriers, and $\frac{1}{4}$ of children will be neither affected nor carriers.

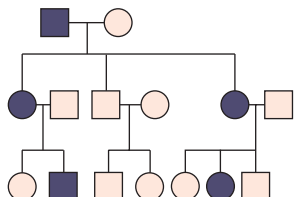
X-linked recessive



Sons of heterozygous mothers have a 50% chance of being affected. No male-to-male transmission. Skips generations.

Commonly more severe in males. Females usually must be homozygous to be affected.

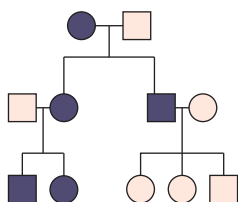
X-linked dominant



Transmitted through both parents. Mothers transmit to 50% of daughters and sons; fathers transmit to all daughters but no sons.

Hypophosphatemic rickets—formerly known as vitamin D-resistant rickets. Inherited disorder resulting in ↑ phosphate wasting at proximal tubule. Results in rickets-like presentation. Other examples: fragile X syndrome, Alport syndrome.

Mitochondrial inheritance



Transmitted only through the mother. All offspring of affected females may show signs of disease.

Variable expression in a population or even within a family due to heteroplasmy.

Mitochondrial myopathies—rare disorders; often present with myopathy, lactic acidosis, and CNS disease, eg, MELAS syndrome (mitochondrial encephalopathy, lactic acidosis, and stroke-like episodes). 2° to failure in oxidative phosphorylation. Muscle biopsy often shows “ragged red fibers” (due to accumulation of diseased mitochondria).

□ = unaffected male; ■ = affected male; ○ = unaffected female; ● = affected female.

Autosomal dominant diseases

Achondroplasia, autosomal dominant polycystic kidney disease, familial adenomatous polyposis, familial hypercholesterolemia, hereditary hemorrhagic telangiectasia, hereditary spherocytosis, Huntington disease, Li-Fraumeni syndrome, Marfan syndrome, multiple endocrine neoplasias, neurofibromatosis type 1 (von Recklinghausen disease), neurofibromatosis type 2, tuberous sclerosis, von Hippel-Lindau disease.

Autosomal recessive diseases

Albinism, autosomal recessive polycystic kidney disease (ARPKD), cystic fibrosis, glycogen storage diseases, hemochromatosis, Kartagener syndrome, mucopolysaccharidoses (except Hunter syndrome), phenylketonuria, sickle cell anemia, sphingolipidoses (except Fabry disease), thalassemias, Wilson disease.

Cystic fibrosis**GENETICS**

Autosomal recessive; defect in *CFTR* gene on chromosome 7; commonly a deletion of Phe508. Most common lethal genetic disease in Caucasian population.

PATHOPHYSIOLOGY

CFTR encodes an ATP-gated Cl^- channel that secretes Cl^- in lungs and GI tract, and reabsorbs Cl^- in sweat glands. Most common mutation → misfolded protein → protein retained in RER and not transported to cell membrane, causing ↓ Cl^- (and H_2O) secretion; ↑ intracellular Cl^- results in compensatory ↑ Na^+ reabsorption via epithelial Na^+ channels → ↑ H_2O reabsorption → abnormally thick mucus secreted into lungs and GI tract. ↑ Na^+ reabsorption also causes more negative transepithelial potential difference.

DIAGNOSIS

↑ Cl^- concentration ($> 60 \text{ mEq/L}$) in sweat is diagnostic. Can present with contraction alkalosis and hypokalemia (ECF effects analogous to a patient taking a loop diuretic) because of ECF $\text{H}_2\text{O}/\text{Na}^+$ losses and concomitant renal K^+/H^+ wasting. ↑ immunoreactive trypsinogen (newborn screening).

COMPLICATIONS

Recurrent pulmonary infections (eg, *S aureus* [early infancy], *P aeruginosa* [adolescence]), chronic bronchitis and bronchiectasis → reticulonodular pattern on CXR, opacification of sinuses. Pancreatic insufficiency, malabsorption with steatorrhea, fat-soluble vitamin deficiencies (A, D, E, K), biliary cirrhosis, liver disease. Meconium ileus in newborns. Infertility in men (absence of vas deferens, spermatogenesis may be unaffected) and subfertility in women (amenorrhea, abnormally thick cervical mucus). Nasal polyps, clubbing of nails.

TREATMENT

Multifactorial: chest physiotherapy, albuterol, aerosolized dornase alfa (DNase), and hypertonic saline facilitate mucus clearance. Azithromycin used as anti-inflammatory agent. Ibuprofen slows disease progression. Pancreatic enzymes for insufficiency.

X-linked recessive disorders

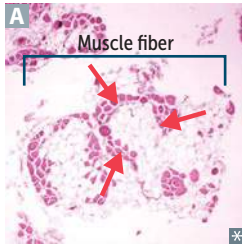
Ornithine transcarbamylase deficiency, Fabry disease, Wiskott-Aldrich syndrome, Ocular albinism, G6PD deficiency, Hunter syndrome, Bruton agammaglobulinemia, Hemophilia A and B, Lesch-Nyhan syndrome, Duchenne (and Becker) muscular dystrophy.

Lyonization—female carriers variably affected depending on the pattern of inactivation of the X chromosome carrying the mutant vs normal gene.

Oblivious Female Will Often Give Her Boys Her x-Linked Disorders

Muscular dystrophies

Duchenne



X-linked disorder typically due to **frameshift** or nonsense mutations → truncated or absent dystrophin protein → progressive myofiber damage. Weakness begins in pelvic girdle muscles and progresses superiorly. Pseudohypertrophy of calf muscles due to fibrofatty replacement of muscle **A**. Waddling gait. Onset before 5 years of age. Dilated cardiomyopathy is common cause of death.

Duchenne = **d**eleted **d**ystrophin.

Dystrophin gene (*DMD*) is the largest protein-coding human gene → ↑ chance of spontaneous mutation. Dystrophin helps anchor muscle fibers, primarily in skeletal and cardiac muscle. It connects the intracellular cytoskeleton (actin) to the transmembrane proteins α - and β -dystroglycan, which are connected to the extracellular matrix (ECM). Loss of dystrophin results in myonecrosis. ↑ CK and aldolase are seen; genetic testing confirms diagnosis.

Becker

X-linked disorder typically due to **non-frameshift** deletions in dystrophin gene (partially functional instead of truncated). Less severe than Duchenne. Onset in adolescence or early adulthood.

Deletions can cause both Duchenne and Becker muscular dystrophies. 2/3 of cases have large deletions spanning one or more exons.

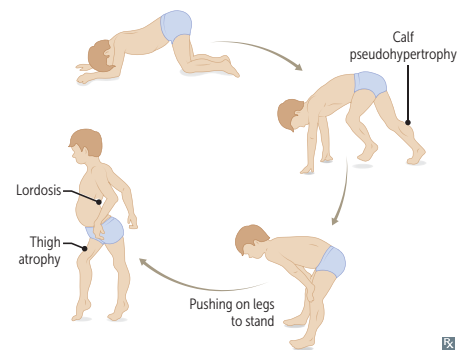
Myotonic type 1

Autosomal dominant. **CTG** trinucleotide repeat expansion in the *DMPK* gene → abnormal expression of myotonin protein kinase → myotonia, muscle wasting, cataracts, testicular atrophy, frontal balding, arrhythmia.

Cataracts, **T**oupee (early balding in men), **G**onadal atrophy.

Gower sign—patient uses upper extremities to help stand up.

Classically seen in Duchenne muscular dystrophy, but also seen in other muscular dystrophies and inflammatory myopathies (eg, polymyositis).



Fragile X syndrome

X-linked dominant inheritance. Trinucleotide repeat in *FMRI* gene → hypermethylation → ↓ expression. Most common cause of inherited intellectual disability and autism and 2nd most common cause of genetically associated mental deficiency (after Down syndrome). Findings: post-pubertal macroorchidism (enlarged testes), long face with a large jaw, large everted ears, autism, mitral valve prolapse.

Trinucleotide repeat disorder (**CGG**)_n.
Chin (protruding), **G**iant **G**onads

Trinucleotide repeat expansion diseases

Huntington disease, **m**yotonic dystrophy, **f**ragile **X** syndrome, and **F**riedreich ataxia.
 May show genetic anticipation (disease severity ↑ and age of onset ↓ in successive generations).
 Huntington disease = (**CAG**)_n
 Myotonic dystrophy = (**CTG**)_n
 Fragile X syndrome = (**CGG**)_n
 Friedreich ataxia = (**GAA**)_n

Try (trinucleotide) **h**unting for **m**y **f**ragile cage-free **e**ggs (**X**).

Caudate has ↓ **A**Ch and **G**ABA
Cataracts, **T**oupee (early balding in men),
Gonadal atrophy
Chin (protruding), **G**iant **G**onads
 Ataxic **GAA**it

Autosomal trisomies**Down syndrome**
(trisomy 21)

Findings: intellectual disability, flat facies, prominent epicanthal folds, single palmar crease, gap between 1st 2 toes, duodenal atresia, Hirschsprung disease, congenital heart disease (eg, atrioventricular septal defect), Brushfield spots. Associated with early-onset Alzheimer disease (chromosome 21 codes for amyloid precursor protein) and ↑ risk of ALL and AML.

95% of cases due to meiotic nondisjunction (↑ with advanced maternal age; from 1:1500 in women < 20 to 1:25 in women > 45 years old).

4% of cases due to unbalanced Robertsonian translocation, most typically between chromosomes 14 and 21. 1% of cases due to mosaicism (no association with maternal nondisjunction; postfertilization mitotic error).

Incidence 1:700.

Drinking age (21).

Most common viable chromosomal disorder and most common cause of genetic intellectual disability.

First-trimester ultrasound commonly shows ↑ nuchal translucency and hypoplastic nasal bone; ↓ serum PAPP-A, ↑ free β-hCG.

Second-trimester quad screen shows ↓ α-fetoprotein, ↑ β-hCG, ↓ estriol, ↑ inhibin A.

Edwards syndrome
(trisomy 18)

Findings: **PRINCE** Edward—**P**rominent occiput, **R**ocker-bottom feet, **I**ntellectual disability, **N**ondisjunction, **C**lenched fists (with overlapping fingers), low-set **E**ars, micrognathia (small jaw), congenital heart disease. Death usually occurs by age 1.

Incidence 1:8000.

Election age (18).

2nd most common autosomal trisomy resulting in live birth (most common is Down syndrome).

PAPP-A and free β-hCG are ↓ in first trimester.

Quad screen shows ↓ α-fetoprotein, ↓ β-hCG, ↓ estriol, ↓ or normal inhibin A.

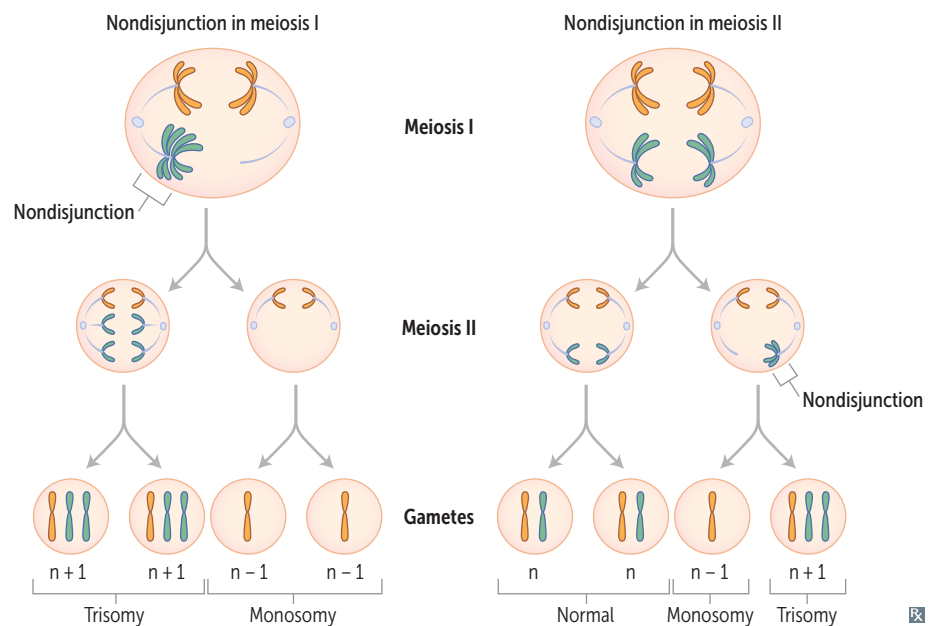
Patau syndrome
(trisomy 13)

Findings: severe intellectual disability, rocker-bottom feet, microphthalmia, microcephaly, cleft li**P**/Palate, holo**P**rosencephaly, **P**olydactyly, cutis **aP**lasia, congenital heart disease. Death usually occurs by age 1.

Incidence 1:15,000.

Puberty (13).

First-trimester pregnancy screen shows ↓ free β-hCG, ↓ PAPP-A.



Genetic disorders by chromosome

CHROMOSOME	SELECTED EXAMPLES
3	von Hippel-Lindau disease, renal cell carcinoma
4	ADPKD (PKD2), achondroplasia, Huntington disease
5	Cri-du-chat syndrome, familial adenomatous polyposis
6	Hemochromatosis (<i>HFE</i>)
7	Williams syndrome, cystic fibrosis
9	Friedreich ataxia
11	Wilms tumor, β -globin gene defects (eg, sickle cell disease, β -thalassemia, MEN1)
13	Patau syndrome, Wilson disease, retinoblastoma (<i>RBI</i>), <i>BRCA2</i>
15	Prader-Willi syndrome, Angelman syndrome, Marfan syndrome
16	ADPKD (<i>PKD1</i>), α -globin gene defects (eg, α -thalassemia)
17	Neurofibromatosis type 1, <i>BRCA1</i> , <i>p53</i>
18	Edwards syndrome
21	Down syndrome
22	Neurofibromatosis type 2, DiGeorge syndrome (22q11)
X	Fragile X syndrome, X-linked agammaglobulinemia, Klinefelter syndrome (XXY)

Robertsonian translocation

Chromosomal translocation that commonly involves chromosome pairs 13, 14, 15, 21, and 22. One of the most common types of translocation. Occurs when the long arms of 2 acrocentric chromosomes (chromosomes with centromeres near their ends) fuse at the centromere and the 2 short arms are lost. Balanced translocations normally do not cause any abnormal phenotype. Unbalanced translocations can result in miscarriage, stillbirth, and chromosomal imbalance (eg, Down syndrome, Patau syndrome).

Cri-du-chat syndrome

Congenital microdeletion of short arm of chromosome 5 (46,XX or XY, 5p[−]). Findings: microcephaly, moderate to severe intellectual disability, high-pitched **crying**/**meowing**, epicanthal folds, cardiac abnormalities (VSD).

Cri du chat = **cry** of the **cat**.

Williams syndrome

Congenital microdeletion of long arm of chromosome 7 (deleted region includes elastin gene). Findings: distinctive “elfin” facies, intellectual disability, hypercalcemia ($\uparrow$ sensitivity to vitamin D), well-developed verbal skills, extreme friendliness with strangers, cardiovascular problems.

22q11 deletion syndromes

Microdeletion at chromosome 22q11 → variable presentations including **C**left palate, **A**bnormal facies, **T**hymic aplasia → T-cell deficiency, **C**ardiac defects, and **H**ypocalcemia 2° to parathyroid aplasia.

DiGeorge syndrome—thymic, parathyroid, and cardiac defects.

Velocardiofacial syndrome—palate, facial, and cardiac defects.

CATCH-22.

Due to aberrant development of 3rd and 4th branchial pouches.

► BIOCHEMISTRY—NUTRITION

Vitamins: fat soluble

A, D, E, K. Absorption dependent on gut and pancreas. Toxicity more common than for water-soluble vitamins because fat-soluble vitamins accumulate in fat.

Malabsorption syndromes with steatorrhea, such as cystic fibrosis and celiac disease, or mineral oil intake can cause fat-soluble vitamin deficiencies.

Vitamins: water soluble

B₁ (thiamine: TPP)
 B₂ (riboflavin: FAD, FMN)
 B₃ (niacin: NAD⁺)
 B₅ (pantothenic acid: CoA)
 B₆ (pyridoxine: PLP)
 B₇ (biotin)
 B₉ (folate)
 B₁₂ (cobalamin)
 C (ascorbic acid)

All wash out easily from body except B₁₂ and B₉ (folate). B₁₂ stored in liver for ~ 3–4 years. B₉ stored in liver for ~ 3–4 months.

B-complex deficiencies often result in dermatitis, glossitis, and diarrhea.

Can be coenzymes (eg, ascorbic acid) or precursors to organic cofactors (eg, FAD, NAD⁺).

Vitamin A (retinol)

FUNCTION	Antioxidant; constituent of visual pigments (retinal); essential for normal differentiation of epithelial cells into specialized tissue (pancreatic cells, mucus-secreting cells); prevents squamous metaplasia. Used to treat measles and acute promyelocytic leukemia (APL).	Retinol is vitamin A , so think retin-A (used topically for wrinkles and A cne). Found in liver and leafy vegetables. Use oral isotretinoin to treat severe cystic acne. Use <i>all-trans</i> retinoic acid to treat acute promyelocytic leukemia.
DEFICIENCY	Night blindness (nyctalopia); dry, scaly skin (xerosis cutis); corneal degeneration (keratomalacia); Bitot spots on conjunctiva; immunosuppression.	
EXCESS	Acute toxicity—nausea, vomiting, vertigo, and blurred vision. Chronic toxicity—alopecia, dry skin (eg, scaliness), hepatic toxicity and enlargement, arthralgias, and pseudotumor cerebri. Teratogenic (cleft palate, cardiac abnormalities), therefore a $\ominus$ pregnancy test and two forms of contraception are required before isotretinoin (vitamin A derivative) is prescribed.	Isotretinoin is teratogenic.

Vitamin B₁ (thiamine)

FUNCTION	In thiamine pyrophosphate (TPP), a cofactor for several dehydrogenase enzyme reactions: - Pyruvate dehydrogenase (links glycolysis to TCA cycle) α-ketoglutarate dehydrogenase (TCA cycle) - Transketolase (HMP shunt) - Branched-chain ketoacid dehydrogenase	Think ATP : α -ketoglutarate dehydrogenase, T ransketolase, and P yruvate dehydrogenase. Spell beriberi as Ber1Ber1 to remember vitamin B₁ . Wernicke-Korsakoff syndrome —confusion, ophthalmoplegia, ataxia (classic triad) + confabulation, personality change, memory loss (permanent). Damage to medial dorsal nucleus of thalamus, mammillary bodies. Dry beriberi —polyneuritis, symmetrical muscle wasting. Wet beriberi —high-output cardiac failure (dilated cardiomyopathy), edema.
DEFICIENCY	Impaired glucose breakdown $\rightarrow$ ATP depletion worsened by glucose infusion; highly aerobic tissues (eg, brain, heart) are affected first. In alcoholic or malnourished patients, give thiamine before dextrose to $\downarrow$ risk of precipitating Wernicke encephalopathy. Diagnosis made by $\uparrow$ in RBC transketolase activity following vitamin B ₁ administration.	

Vitamin B₂ (riboflavin)

FUNCTION	Component of flavins FAD and FMN, used as cofactors in redox reactions, eg, the succinate dehydrogenase reaction in the TCA cycle.	FAD and FMN are derived from ribo F lavin (B ₂ ≈ 2 ATP).
DEFICIENCY	Cheilosis (inflammation of lips, scaling and fissures at the corners of the mouth), Corneal vascularization.	The 2 C 's of B ₂ .

Vitamin B₃ (niacin)

FUNCTION	Constituent of NAD ⁺ , NADP ⁺ (used in redox reactions). Derived from tryptophan. Synthesis requires vitamins B ₂ and B ₆ . Used to treat dyslipidemia; lowers levels of VLDL and raises levels of HDL.	NAD derived from N iacin (B ₃ ≈ 3 ATP).
DEFICIENCY	Glossitis. Severe deficiency leads to pellagra, which can be caused by Hartnup disease, malignant carcinoid syndrome (↑ tryptophan metabolism), and isoniazid (↓ vitamin B ₆). Symptoms of pellagra: D iarrhea, D ementia (also hallucinations), D ermatitis (C3/C4 dermatome circumferential “broad collar” rash [Casal necklace], hyperpigmentation of sun-exposed limbs A).	The 3 D 's of B ₃ . Hartnup disease —autosomal recessive. Deficiency of neutral amino acid (eg, tryptophan) transporters in proximal renal tubular cells and on enterocytes → neutral aminoaciduria and ↓ absorption from the gut → ↓ tryptophan for conversion to niacin → pellagra-like symptoms. Treat with high-protein diet and nicotinic acid.
EXCESS	Facial flushing (induced by prostaglandin, not histamine; can avoid by taking aspirin with niacin), hyperglycemia, hyperuricemia.	

**Vitamin B₅ (pantothenic acid)**

FUNCTION	Essential component of coenzyme A (CoA, a cofactor for acyl transfers) and fatty acid synthase.	B ₅ is “ pento ”thenic acid.
DEFICIENCY	Dermatitis, enteritis, alopecia, adrenal insufficiency.	

Vitamin B₆ (pyridoxine)

FUNCTION	Converted to pyridoxal phosphate (PLP), a cofactor used in transamination (eg, ALT and AST), decarboxylation reactions, glycogen phosphorylase. Synthesis of cystathionine, heme, niacin, histamine, and neurotransmitters including serotonin, epinephrine, norepinephrine (NE), dopamine, and GABA.	
DEFICIENCY	Convulsions, hyperirritability, peripheral neuropathy (deficiency inducible by isoniazid and oral contraceptives), sideroblastic anemias due to impaired hemoglobin synthesis and iron excess.	

Vitamin B₇ (biotin)

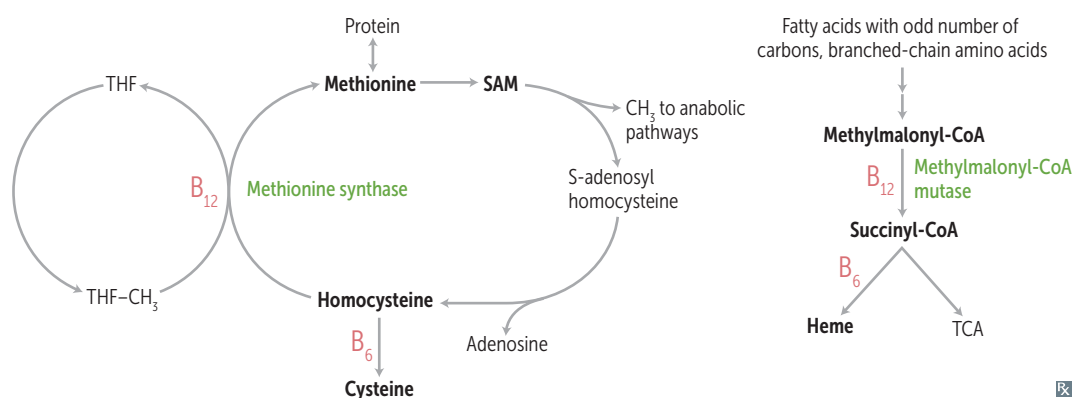
FUNCTION	Cofactor for carboxylation enzymes (which add a 1-carbon group): - Pyruvate carboxylase: pyruvate (3C) → oxaloacetate (4C) - Acetyl-CoA carboxylase: acetyl-CoA (2C) → malonyl-CoA (3C) - Propionyl-CoA carboxylase: propionyl-CoA (3C) → methylmalonyl-CoA (4C)	“ A vidin in egg whites a vidly binds biotin.”
DEFICIENCY	Relatively rare. Dermatitis, alopecia, enteritis. Caused by antibiotic use or excessive ingestion of raw egg whites.	

Vitamin B₉ (folate)

FUNCTION	Converted to tetrahydrofolic acid (THF), a coenzyme for 1-carbon transfer/methylation reactions. Important for the synthesis of nitrogenous bases in DNA and RNA.	Found in leafy green vegetables. Absorbed in jejunum. Folate from foliage. Small reserve pool stored primarily in the liver.
DEFICIENCY	Macrocytic, megaloblastic anemia; hypersegmented polymorphonuclear cells (PMNs); glossitis; no neurologic symptoms (as opposed to vitamin B₁₂ deficiency). Labs: ↑ homocysteine, normal methylmalonic acid levels. Most common vitamin deficiency in the United States. Seen in alcoholism and pregnancy.	Deficiency can be caused by several drugs (eg, phenytoin, sulfonamides, methotrexate). Supplemental maternal folic acid at least 1 month prior to conception and during early pregnancy to ↓ risk of neural tube defects.

Vitamin B₁₂ (cobalamin)

FUNCTION	Cofactor for methionine synthase (transfers CH ₃ groups as methylcobalamin) and methylmalonyl-CoA mutase. Important for DNA synthesis.	Found in animal products. Synthesized only by microorganisms. Very large reserve pool (several years) stored primarily in the liver. Deficiency caused by malabsorption (eg, sprue, enteritis, <i>Diphyllobothrium latum</i>), lack of intrinsic factor (pernicious anemia, gastric bypass surgery), absence of terminal ileum (surgical resection, eg, for Crohn disease), or insufficient intake (eg, veganism). Anti-intrinsic factor antibodies diagnostic for pernicious anemia.
DEFICIENCY	Macrocytic, megaloblastic anemia; hypersegmented PMNs; paresthesias and subacute combined degeneration (degeneration of dorsal columns, lateral corticospinal tracts, and spinocerebellar tracts) due to abnormal myelin. Associated with ↑ serum homocysteine and methylmalonic acid levels, along with 2° folate deficiency. Prolonged deficiency → irreversible nerve damage.	

**Vitamin C (ascorbic acid)**

FUNCTION	Antioxidant; also facilitates iron absorption by reducing it to Fe ²⁺ state. Necessary for hydroxylation of proline and lysine in collagen synthesis. Necessary for dopamine β-hydroxylase, which converts dopamine to NE.	Found in fruits and vegetables. Pronounce “ absorbic ” acid. Ancillary treatment for methemoglobinemia by reducing Fe ³⁺ to Fe ²⁺ .
DEFICIENCY	Scurvy —swollen gums, bruising, petechiae, hemarthrosis, anemia, poor wound healing, perifollicular and subperiosteal hemorrhages, “corkscrew” hair. Weakened immune response.	Vitamin C deficiency causes sCurvy due to a C ollagen synthesis defect.
EXCESS	Nausea, vomiting, diarrhea, fatigue, calcium oxalate nephrolithiasis. Can ↑ risk of iron toxicity in predisposed individuals (eg, those with transfusions, hereditary hemochromatosis).	

Vitamin D

D₂ = ergocalciferol—ingested from plants.

D₃ = cholecalciferol—consumed in milk, formed in sun-exposed skin (stratum basale).

25-OH D₃ = storage form.

1,25-(OH)₂ D₃ (calcitriol) = active form.

FUNCTION

↑ intestinal absorption of calcium and phosphate, ↑ bone mineralization at low levels, ↑ bone resorption at higher levels.

DEFICIENCY

Rickets in children (deformity, such as genu varum “bow legs” **A**), osteomalacia in adults (bone pain and muscle weakness), hypocalcemic tetany. Breastfed infants should receive oral vitamin D. Deficiency is exacerbated by low sun exposure, pigmented skin, prematurity.

**EXCESS**

Hypercalcemia, hypercalciuria, loss of appetite, stupor. Seen in granulomatous disease (↑ activation of vitamin D by epithelioid macrophages).

Vitamin E (tocopherol/tocotrienol)**FUNCTION**

Antioxidant (protects RBCs and membranes from free radical damage).

High-dose supplementation may alter metabolism of vitamin K → enhanced anticoagulant effects of warfarin.

DEFICIENCY

Hemolytic anemia, acanthocytosis, muscle weakness, posterior column and spinocerebellar tract demyelination.

Neurologic presentation may appear similar to vitamin B₁₂ deficiency, but without megaloblastic anemia, hypersegmented neutrophils, or ↑ serum methylmalonic acid levels.

Vitamin K (phytomenadione, phylloquinone, phytonadione)**FUNCTION**

Activated by epoxide reductase to the reduced form, which is a cofactor for the γ-carboxylation of glutamic acid residues on various proteins required for blood clotting. Synthesized by intestinal flora.

K is for **K**oagulation. Necessary for the maturation of clotting factors II, VII, IX, X, and proteins C and S. Warfarin inhibits vitamin K–dependent synthesis of these factors and proteins.

DEFICIENCY

Neonatal hemorrhage with ↑ PT and ↑ aPTT but normal bleeding time (neonates have sterile intestines and are unable to synthesize vitamin K). Can also occur after prolonged use of broad-spectrum antibiotics.

Not in breast milk; neonates are given vitamin K injection at birth to prevent hemorrhagic disease of the newborn.

Zinc**FUNCTION**

Mineral essential for the activity of 100+ enzymes. Important in the formation of zinc fingers (transcription factor motif).

DEFICIENCY

Delayed wound healing, hypogonadism, ↓ adult hair (axillary, facial, pubic), dysgeusia, anosmia, acrodermatitis enteropathica **A**. May predispose to alcoholic cirrhosis.

**Protein-energy malnutrition****Kwashiorkor**

Protein malnutrition resulting in skin lesions, edema due to ↓ plasma oncotic pressure, liver malfunction (fatty change due to ↓ apolipoprotein synthesis). Clinical picture is small child with swollen abdomen **A**.

Kwashiorkor results from protein-deficient **MEALS**:

Malnutrition

Edema

Anemia

Liver (fatty)

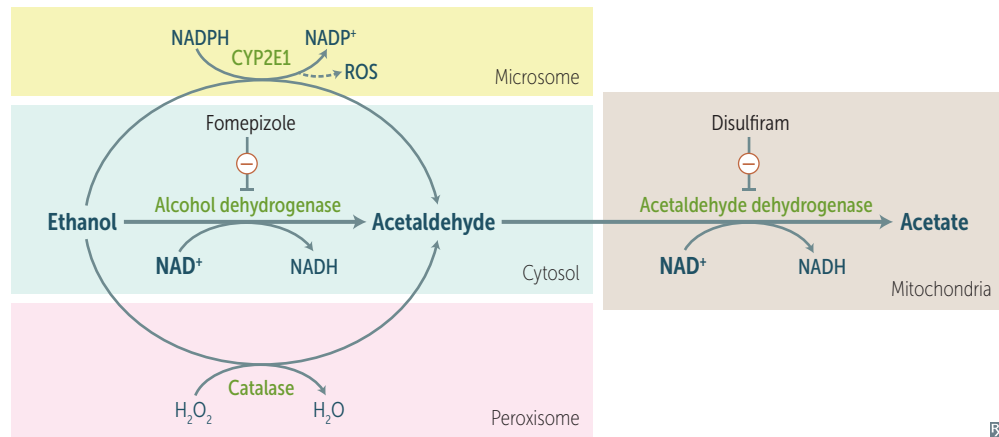
Skin lesions (hyperkeratosis/hyperpigmentation)

**Marasmus**

Malnutrition not causing edema. Diet is deficient in calories but no nutrients are entirely absent.

Marasmus results in **M**uscle wasting **B**.

Ethanol metabolism



FOMEpizole—inhibits alcohol dehydrogenase and is an antidote **F**or **O**verdoses of **M**ethanol or **E**thylene glycol.

Disulfiram—inhibits acetaldehyde dehydrogenase (acetaldehyde accumulates, contributing to hangover symptoms), **d**iscouraging drinking.

NAD⁺ is the limiting reagent.

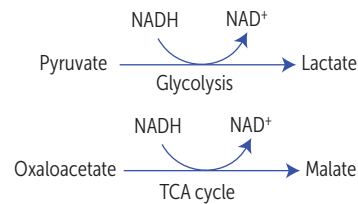
Alcohol dehydrogenase operates via zero-order kinetics.

Ethanol metabolism ↑ NADH/NAD⁺ ratio in liver, causing:

- Pyruvate → lactate (lactic acidosis)
- Oxaloacetate → malate (prevents gluconeogenesis → fasting hypoglycemia)
- Dihydroxyacetone phosphate → glycerol-3-phosphate (combines with fatty acids to make triglycerides → hepatosteatosis)

End result is clinical picture seen in chronic alcoholism.

Additionally, ↑ NADH/NAD⁺ ratio disfavors TCA production of NADH → ↑ utilization of acetyl-CoA for ketogenesis (→ ketoacidosis) and lipogenesis (→ hepatosteatosis).



► BIOCHEMISTRY—METABOLISM

Metabolism sites

Mitochondria

Fatty acid oxidation (β-oxidation), acetyl-CoA production, TCA cycle, oxidative phosphorylation, ketogenesis.

Cytoplasm

Glycolysis, HMP shunt, and synthesis of steroids (SER), proteins (ribosomes, RER), fatty acids, cholesterol, and nucleotides.

Both

Heme synthesis, **U**rea cycle, **G**luconeogenesis. **HUGs** take **two** (ie, both).

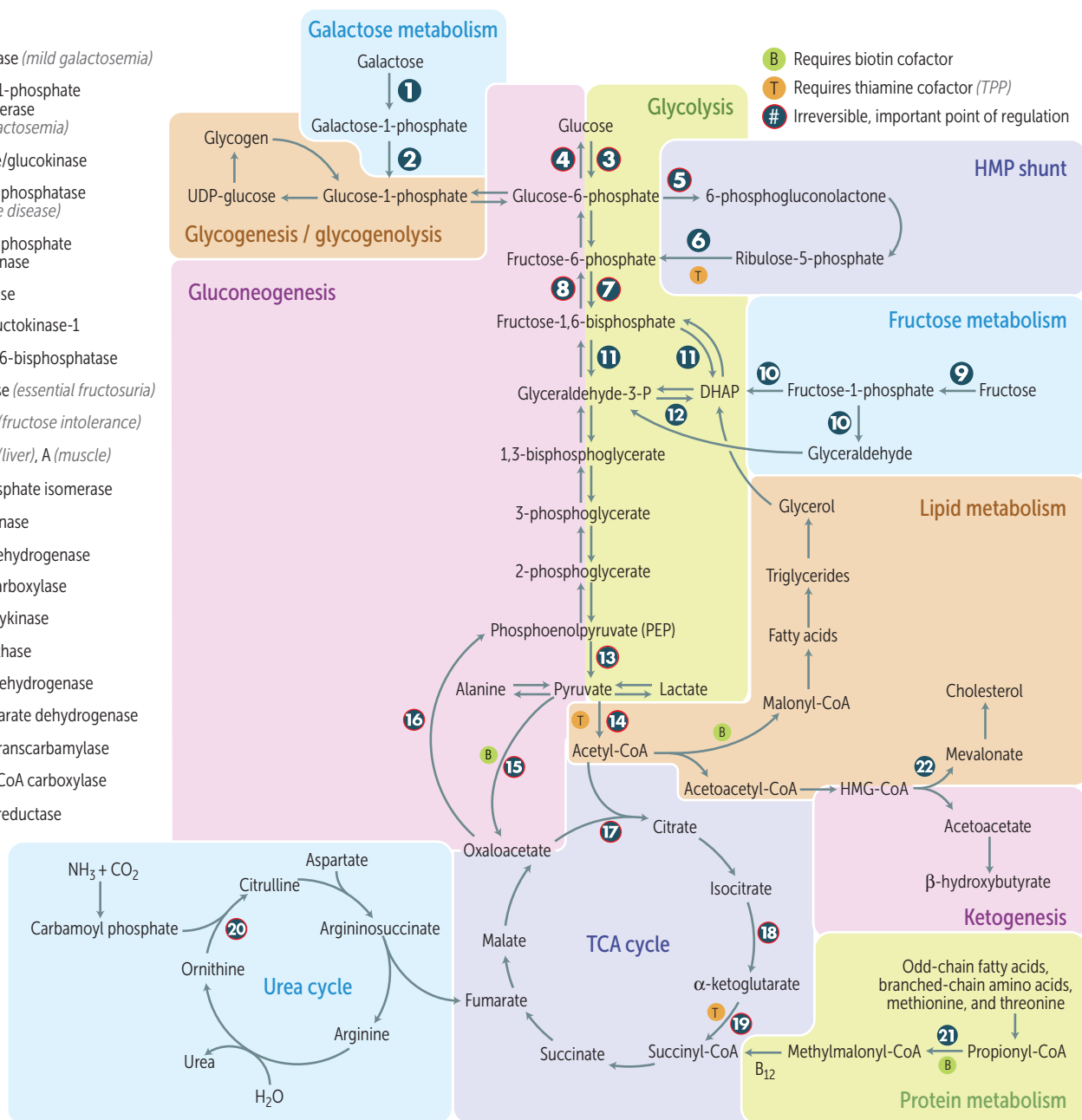
Enzyme terminology	An enzyme's name often describes its function. For example, glucokinase is an enzyme that catalyzes the phosphorylation of glucose using a molecule of ATP. The following are commonly used enzyme descriptors.
Kinase	Catalyzes transfer of a phosphate group from a high-energy molecule (usually ATP) to a substrate (eg, phosphofructokinase).
Phosphorylase	Adds inorganic phosphate onto substrate without using ATP (eg, glycogen phosphorylase).
Phosphatase	Removes phosphate group from substrate (eg, fructose-1,6-bisphosphatase).
Dehydrogenase	Catalyzes oxidation-reduction reactions (eg, pyruvate dehydrogenase).
Hydroxylase	Adds hydroxyl group (–OH) onto substrate (eg, tyrosine hydroxylase).
Carboxylase	Transfers CO ₂ groups with the help of biotin (eg, pyruvate carboxylase).
Mutase	Relocates a functional group within a molecule (eg, vitamin B ₁₂ –dependent methylmalonyl-CoA mutase).
Synthase/synthetase	Combines 2 molecules into 1 (condensation reaction) either using an energy source (synthase, eg, glycogen synthase) or not (synthetase, eg, PRPP synthetase).

Rate-determining enzymes of metabolic processes

PROCESS	ENZYME	REGULATORS
Glycolysis	Phosphofructokinase-1 (PFK-1)	AMP ⊕, fructose-2,6-bisphosphate ⊕ ATP ⊖, citrate ⊖
Gluconeogenesis	Fructose-1,6-bisphosphatase	Citrate ⊕ AMP ⊖, fructose-2,6-bisphosphate ⊖
TCA cycle	Isocitrate dehydrogenase	ADP ⊕ ATP ⊖, NADH ⊖
Glycogenesis	Glycogen synthase	Glucose-6-phosphate ⊕, insulin ⊕, cortisol ⊕ Epinephrine ⊖, glucagon ⊖
Glycogenolysis	Glycogen phosphorylase	Epinephrine ⊕, glucagon ⊕, AMP ⊕ Glucose-6-phosphate ⊖, insulin ⊖, ATP ⊖
HMP shunt	Glucose-6-phosphate dehydrogenase (G6PD)	NADP ⁺ ⊕ NADPH ⊖
De novo pyrimidine synthesis	Carbamoyl phosphate synthetase II	ATP ⊕, PRPP ⊕ UTP ⊖
De novo purine synthesis	Glutamine-phosphoribosylpyrophosphate (PRPP) amidotransferase	AMP ⊖, inosine monophosphate (IMP) ⊖, GMP ⊖
Urea cycle	Carbamoyl phosphate synthetase I	N-acetylglutamate ⊕
Fatty acid synthesis	Acetyl-CoA carboxylase (ACC)	Insulin ⊕, citrate ⊕ Glucagon ⊖, palmitoyl-CoA ⊖
Fatty acid oxidation	Carnitine acyltransferase I	Malonyl-CoA ⊖
Ketogenesis	HMG-CoA synthase	
Cholesterol synthesis	HMG-CoA reductase	Insulin ⊕, thyroxine ⊕ Glucagon ⊖, cholesterol ⊖

Summary of pathways

- 1 Galactokinase (*mild galactosemia*)
- 2 Galactose-1-phosphate uridylyltransferase (*severe galactosemia*)
- 3 Hexokinase/glucokinase
- 4 Glucose-6-phosphatase (*von Gierke disease*)
- 5 Glucose-6-phosphate dehydrogenase
- 6 Transketolase
- 7 Phosphofructokinase-1
- 8 Fructose-1,6-bisphosphatase
- 9 Fructokinase (*essential fructosuria*)
- 10 Aldolase B (*fructose intolerance*)
- 11 Aldolase B (*liver*), A (*muscle*)
- 12 Triose phosphate isomerase
- 13 Pyruvate kinase
- 14 Pyruvate dehydrogenase
- 15 Pyruvate carboxylase
- 16 PEP carboxykinase
- 17 Citrate synthase
- 18 Isocitrate dehydrogenase
- 19 α -ketoglutarate dehydrogenase
- 20 Ornithine transcarbamylase
- 21 Propionyl-CoA carboxylase
- 22 HMG-CoA reductase



ATP production

Aerobic metabolism of glucose produces 32 net ATP via malate-aspartate shuttle (heart and liver), 30 net ATP via glycerol-3-phosphate shuttle (muscle).

Anaerobic glycolysis produces only 2 net ATP per glucose molecule.

ATP hydrolysis can be coupled to energetically unfavorable reactions.

Arsenic causes glycolysis to produce zero net ATP.

Activated carriers

CARRIER MOLECULE	CARRIED IN ACTIVATED FORM
ATP	Phosphoryl groups
NADH, NADPH, FADH ₂	Electrons
CoA, lipoamide	Acyl groups
Biotin	CO ₂
Tetrahydrofolates	1-carbon units
S-adenosylmethionine (SAM)	CH ₃ groups
TPP	Aldehydes

Universal electron acceptors

Nicotinamides (NAD⁺ from vitamin B₃, NADP⁺) and flavin nucleotides (FAD⁺ from vitamin B₂).

NAD⁺ is generally used in **catabolic** processes to carry reducing equivalents away as NADH.

NADPH is used in **anabolic** processes (steroid and fatty acid synthesis) as a supply of reducing equivalents.

NADPH is a product of the HMP shunt.

NADPH is used in:

- Anabolic processes
- Respiratory burst
- Cytochrome P-450 system
- Glutathione reductase

Hexokinase vs glucokinase

Phosphorylation of glucose to yield glucose-6-phosphate serves as the 1st committed step of glycolysis (also serves as the 1st step of glycogen synthesis in the liver). Reaction is catalyzed by either hexokinase or glucokinase, depending on the tissue. At low glucose concentrations, hexokinase sequesters glucose in the tissue. At high glucose concentrations, excess glucose is stored in the liver.

	Hexokinase	Glucokinase
Location	Most tissues, except liver and pancreatic β cells	Liver, β cells of pancreas
K _m	Lower ($\uparrow$ affinity)	Higher ($\downarrow$ affinity)
V _{max}	Lower ($\downarrow$ capacity)	Higher ($\uparrow$ capacity)
Induced by insulin	No	Yes
Feedback-inhibited by glucose-6-phosphate	Yes	No

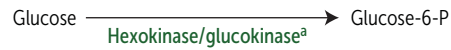
Glycolysis regulation, key enzymes

Net glycolysis (cytoplasm):



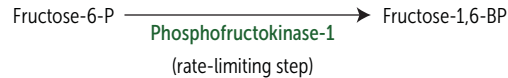
Equation not balanced chemically, and exact balanced equation depends on ionization state of reactants and products.

REQUIRE ATP



Glucose-6-P $\ominus$ hexokinase.

Fructose-6-P $\ominus$ glucokinase.

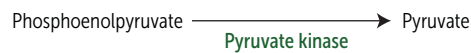
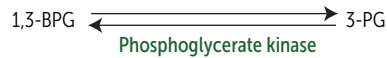


AMP $\oplus$, fructose-2,6-bisphosphate $\oplus$.

ATP $\ominus$, citrate $\ominus$.

^aGlucokinase in liver and β cells of pancreas; hexokinase in all other tissues.

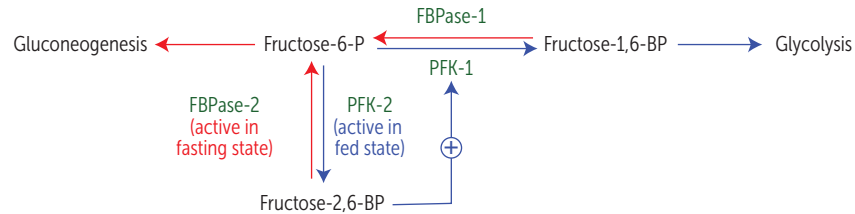
PRODUCE ATP



Fructose-1,6-bisphosphate $\oplus$.

ATP $\ominus$, alanine $\ominus$.

Regulation by fructose-2,6-bisphosphate



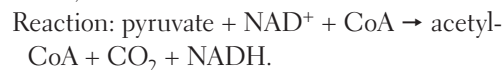
FBPase-2 (fructose bisphosphatase-2) and **PFK-2 (phosphofructokinase-2)** are the same bifunctional enzyme whose function is reversed by phosphorylation by protein kinase A.

Fasting state: $\uparrow$ glucagon $\rightarrow \uparrow$ cAMP $\rightarrow \uparrow$ protein kinase A $\rightarrow \uparrow$ FBPase-2, $\downarrow$ PFK-2, less glycolysis, more gluconeogenesis.

Fed state: $\uparrow$ insulin $\rightarrow \downarrow$ cAMP $\rightarrow \downarrow$ protein kinase A $\rightarrow \downarrow$ FBPase-2, $\uparrow$ PFK-2, more glycolysis, less gluconeogenesis.

Pyruvate dehydrogenase complex

Mitochondrial enzyme complex linking glycolysis and TCA cycle. Differentially regulated in fed/fasting states (active in fed state).



The complex contains 3 enzymes that require 5 cofactors:

1. **T**hiamine pyrophosphate (B_1)
2. **L**ipoic acid
3. **C**oA (B_5 , pantothenic acid)
4. **F**AD (B_2 , riboflavin)
5. **N**AD $^+$ (B_3 , niacin)

Activated by:

- $\uparrow$ NAD $^+$ /NADH ratio
- $\uparrow$ ADP
- $\uparrow$ Ca $^{2+}$

The complex is similar to the α -ketoglutarate dehydrogenase complex (same cofactors, similar substrate and action), which converts α -ketoglutarate $\rightarrow$ succinyl-CoA (TCA cycle).

The Lovely Co-enzymes For Nerds.

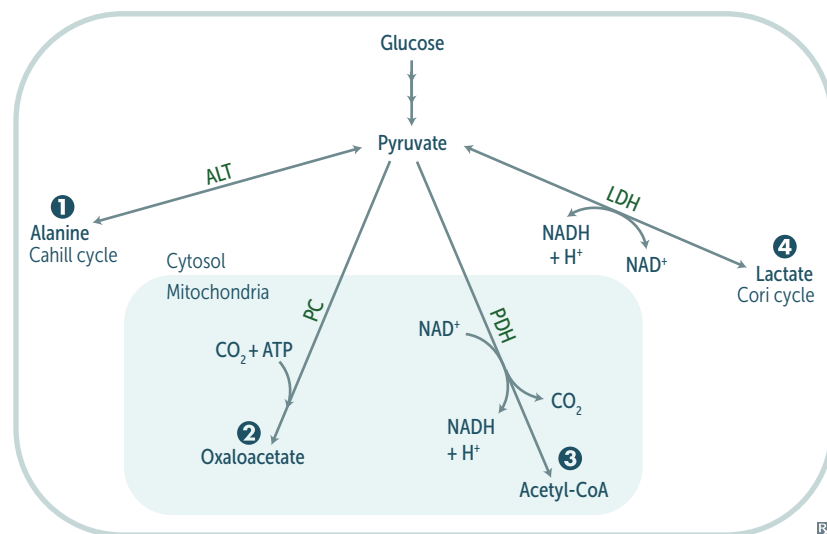
Arsenic inhibits lipoic acid. Arsenic poisoning clinical findings: vomiting, rice-water stools, garlic breath, QT prolongation.

Pyruvate dehydrogenase complex deficiency

Causes a buildup of pyruvate that gets shunted to lactate (via LDH) and alanine (via ALT).
X-linked.

FINDINGS	Neurologic defects, lactic acidosis, ↑ serum alanine starting in infancy.
TREATMENT	↑ intake of ketogenic nutrients (eg, high fat content or ↑ lysine and leucine). Lysine and Leucine—the onLy pureLy ketogenic amino acids.

Pyruvate metabolism

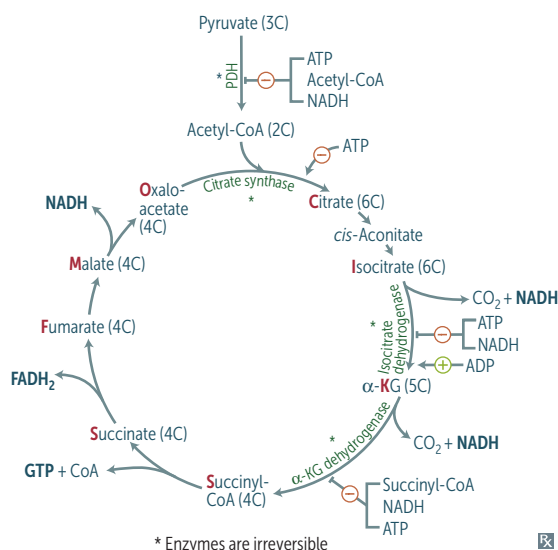


Functions of different pyruvate metabolic pathways (and their associated cofactors):

- 1 Alanine aminotransferase (B₆): alanine carries amino groups to the liver from muscle
- 2 Pyruvate carboxylase (biotin): oxaloacetate can replenish TCA cycle or be used in gluconeogenesis
- 3 Pyruvate dehydrogenase (B₁, B₂, B₃, B₅, lipoic acid): transition from glycolysis to the TCA cycle
- 4 Lactic acid dehydrogenase (B₃): end of anaerobic glycolysis (major pathway in RBCs, WBCs, kidney medulla, lens, testes, and cornea)

TCA cycle (Krebs cycle)

Pyruvate → acetyl-CoA produces 1 NADH, 1 CO₂.



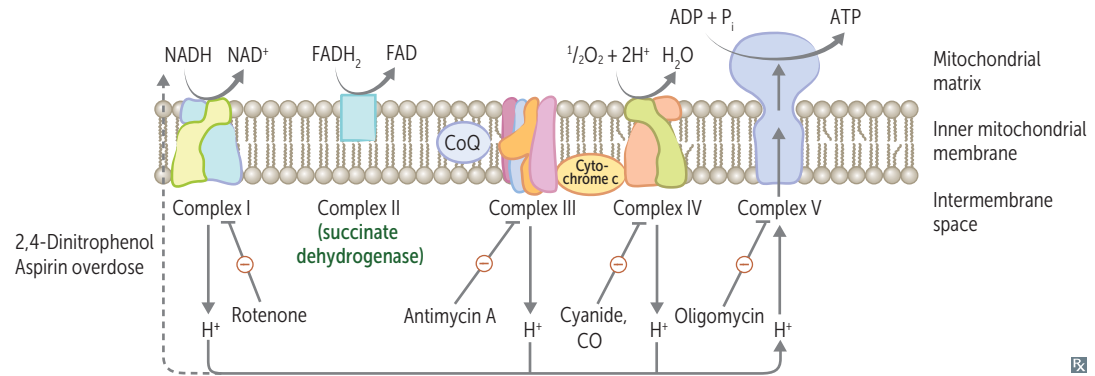
The TCA cycle produces 3 NADH, 1 FADH₂, 2 CO₂, 1 GTP per acetyl-CoA = 10 ATP/ acetyl-CoA (2× everything per glucose). TCA cycle reactions occur in the mitochondria.

α-ketoglutarate dehydrogenase complex requires the same cofactors as the pyruvate dehydrogenase complex (B₁, B₂, B₃, B₅, lipoic acid).

Citrate Is Krebs' Starting Substrate For Making Oxaloacetate.

Electron transport chain and oxidative phosphorylation

NADH electrons from glycolysis enter mitochondria via the malate-aspartate or glycerol-3-phosphate shuttle. FADH_2 electrons are transferred to complex II (at a lower energy level than NADH). The passage of electrons results in the formation of a proton gradient that, coupled to oxidative phosphorylation, drives the production of ATP.



ATP PRODUCED VIA ATP SYNTHASE

1 NADH $\rightarrow$ 2.5 ATP; 1 $\text{FADH}_2 \rightarrow$ 1.5 ATP.

OXIDATIVE PHOSPHORYLATION POISONS

Electron transport inhibitors

Directly inhibit electron transport, causing a $\downarrow$ proton gradient and block of ATP synthesis.

Rotenone: complex **one** inhibitor.
“An-3-mycin” (antimycin) **A**: complex **3** inhibitor.
CO/CN: complex **4** inhibitors (**4** letters).

ATP synthase inhibitors

Directly inhibit mitochondrial ATP synthase, causing an $\uparrow$ proton gradient. No ATP is produced because electron transport stops.

Oligomycin.

Uncoupling agents

$\uparrow$ permeability of membrane, causing a $\downarrow$ proton gradient and $\uparrow$ O_2 consumption. ATP synthesis stops, but electron transport continues. Produces heat.

2,4-Dinitrophenol (used illicitly for weight loss), aspirin (fevers often occur after aspirin overdose), thermogenin in brown fat.

Gluconeogenesis, irreversible enzymes

Pyruvate carboxylase

In mitochondria. Pyruvate $\rightarrow$ oxaloacetate.

Pathway Produces Fresh Glucose.

Phosphoenolpyruvate carboxykinase

In cytosol. Oxaloacetate $\rightarrow$ phosphoenolpyruvate.

Requires biotin, ATP. Activated by acetyl-CoA.

Requires GTP.

Fructose-1,6-bisphosphatase

In cytosol. Fructose-1,6-bisphosphate $\rightarrow$ fructose-6-phosphate.

Citrate $\oplus$, AMP $\ominus$, fructose 2,6-bisphosphate $\ominus$.

Glucose-6-phosphatase

In ER. Glucose-6-phosphate $\rightarrow$ glucose.

Occurs primarily in liver; serves to maintain euglycemia during fasting. Enzymes also found in kidney, intestinal epithelium. Deficiency of the key gluconeogenic enzymes causes hypoglycemia. (Muscle cannot participate in gluconeogenesis because it lacks glucose-6-phosphatase). Odd-chain fatty acids yield 1 propionyl-CoA during metabolism, which can enter the TCA cycle (as succinyl-CoA), undergo gluconeogenesis, and serve as a glucose source. Even-chain fatty acids cannot produce new glucose, since they yield only acetyl-CoA equivalents.

HMP shunt (pentose phosphate pathway)

Provides a source of NADPH from abundantly available glucose-6-P (NADPH is required for reductive reactions, eg, glutathione reduction inside RBCs, fatty acid and cholesterol biosynthesis). Additionally, this pathway yields ribose for nucleotide synthesis and glycolytic intermediates. 2 distinct phases (oxidative and nonoxidative), both of which occur in the cytoplasm. No ATP is used or produced.

Sites: lactating mammary glands, liver, adrenal cortex (sites of fatty acid or steroid synthesis), RBCs.

REACTIONS	KEY ENZYMES	PRODUCTS
Oxidative (irreversible)	Glucose-6-P $\xrightarrow[\text{NADP}^+]{\text{NADPH}}$ Glucose-6-P dehydrogenase $\rightarrow$ CO_2 Rate-limiting step	2 NADPH Ribulose-5-P_i
Nonoxidative (reversible)	Phosphopentose isomerase, transketolases Requires B₁	Ribose-5-P Glyceraldehyde-3-phosphate Fructose-6-P

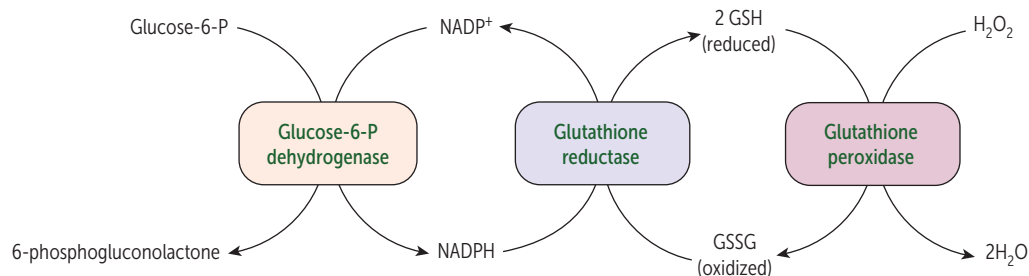
Glucose-6-phosphate dehydrogenase deficiency

NADPH is necessary to keep glutathione reduced, which in turn detoxifies free radicals and peroxides. ↓ NADPH in RBCs leads to hemolytic anemia due to poor RBC defense against oxidizing agents (eg, fava beans, sulfonamides, primaquine, antituberculosis drugs). Infection (most common cause) can also precipitate hemolysis; inflammatory response produces free radicals that diffuse into RBCs, causing oxidative damage.

X-linked recessive disorder; most common human enzyme deficiency; more prevalent among African Americans. ↑ malarial resistance.

Heinz bodies—denatured **H**emoglobin precipitates within RBCs due to oxidative stress.

Bite cells—result from the phagocytic removal of **H**einz bodies by splenic macrophages. Think, “**B**ite into some **H**einz ketchup.”



Disorders of fructose metabolism

Essential fructosuria

Involves a defect in **fructokinase**. Autosomal recessive. A benign, asymptomatic condition, since fructose is not trapped in cells.

Symptoms: fructose appears in blood and urine.

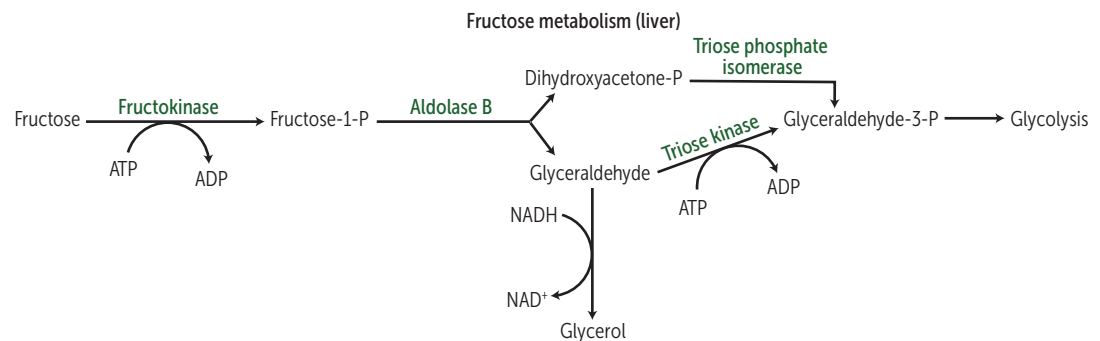
Disorders of fructose metabolism cause milder symptoms than analogous disorders of galactose metabolism.

Fructose intolerance

Hereditary deficiency of **aldolase B**. Autosomal recessive. Fructose-1-phosphate accumulates, causing a ↓ in available phosphate, which results in inhibition of glycogenolysis and gluconeogenesis. Symptoms present following consumption of fruit, juice, or honey. Urine dipstick will be ⊖ (tests for glucose only); reducing sugar can be detected in the urine (nonspecific test for inborn errors of carbohydrate metabolism).

Symptoms: hypoglycemia, jaundice, cirrhosis, vomiting.

Treatment: ↓ intake of both fructose and sucrose (glucose + fructose).



Disorders of galactose metabolism

Galactokinase deficiency

Hereditary deficiency of **galactokinase**. Galactitol accumulates if galactose is present in diet.

Relatively mild condition. Autosomal recessive.

Symptoms: galactose appears in blood (galactosemia) and urine (galactosuria); infantile cataracts.

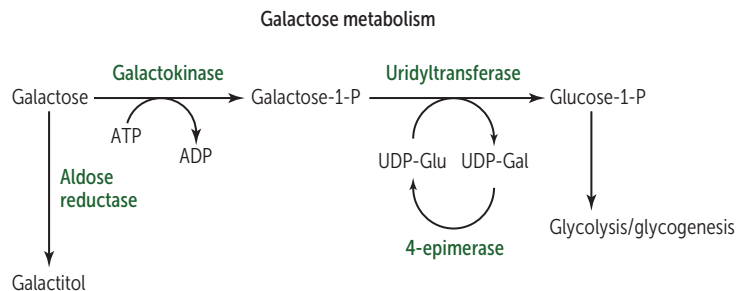
May present as failure to track objects or to develop a social smile.

Classic galactosemia

Absence of **galactose-1-phosphate uridylyltransferase**. Autosomal recessive. Damage is caused by accumulation of toxic substances (including galactitol, which accumulates in the lens of the eye).

Symptoms develop when infant begins feeding (lactose present in breast milk and routine formula) and include failure to thrive, jaundice, hepatomegaly, infantile cataracts, intellectual disability. Can predispose to *E coli* sepsis in neonates.

Treatment: exclude galactose and lactose (galactose + glucose) from diet.



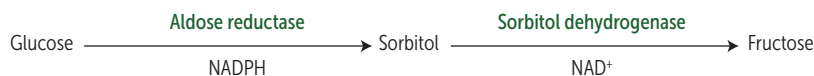
Fructose is to **Aldolase B** as Galactose is to **UridylTransferase (FAB GUT)**.

The more serious defects lead to PO_4^{3-} depletion.

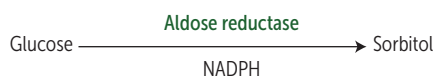
Sorbitol

An alternative method of trapping glucose in the cell is to convert it to its alcohol counterpart, called sorbitol, via aldose reductase. Some tissues then convert sorbitol to fructose using sorbitol dehydrogenase; tissues with an insufficient amount/activity of this enzyme are at risk for intracellular sorbitol accumulation, causing osmotic damage (eg, cataracts, retinopathy, and peripheral neuropathy seen with chronic hyperglycemia in diabetes). High blood levels of galactose also result in conversion to the osmotically active galactitol via aldose reductase.

Liver, ovaries, and seminal vesicles have both enzymes.



Lens has primarily aldose reductase. Retina, Kidneys, and Schwann cells have only aldose reductase (LuRKS).

**Lactase deficiency**

Insufficient lactase enzyme → dietary lactose intolerance. Lactase functions on the brush border to digest lactose (in human and cow milk) into glucose and galactose.
 Primary: age-dependent decline after childhood (absence of lactase-persistent allele), common in people of Asian, African, or Native American descent.
 Secondary: loss of brush border due to gastroenteritis (eg, rotavirus), autoimmune disease, etc.
 Congenital lactase deficiency: rare, due to defective gene.
 Stool demonstrates ↓ pH and breath shows ↑ hydrogen content with lactose hydrogen breath test.
 Intestinal biopsy reveals normal mucosa in patients with hereditary lactose intolerance.

FINDINGS

Bloating, cramps, flatulence, osmotic diarrhea.

TREATMENT

Avoid dairy products or add lactase pills to diet; lactose-free milk.

Amino acids

Only L-amino acids are found in proteins.

Essential

Glucogenic: methionine (**Met**), histidine (**His**), valine (**Val**).

Glucogenic/ketogenic: isoleucine (Ile), phenylalanine (Phe), threonine (Thr), tryptophan (Trp).

Ketogenic: leucine (Leu), lysine (Lys).

I **met his** valentine, she is so **sweet** (glucogenic).
 All essential amino acids need to be supplied in the diet.

Acidic

Aspartic acid (Asp) and glutamic acid (Glu).
 Negatively charged at body pH.

Basic

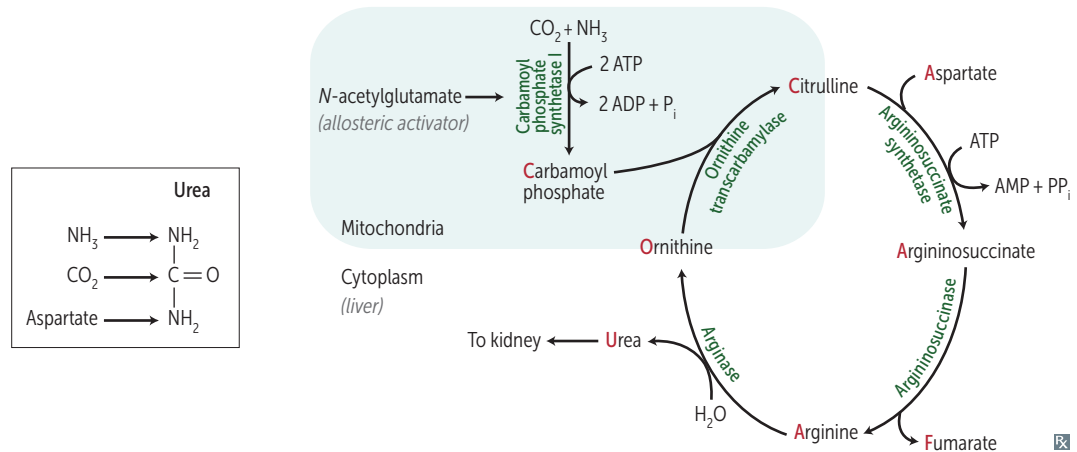
Histidine (**His**), lysine (**Lys**), arginine (**Arg**).
 Arg is most **basic**.
 His has no charge at body pH.

His lys (lies) **are basic**.
 Arg and His are required during periods of growth. Arg and Lys are ↑ in histones, which bind negatively charged DNA.

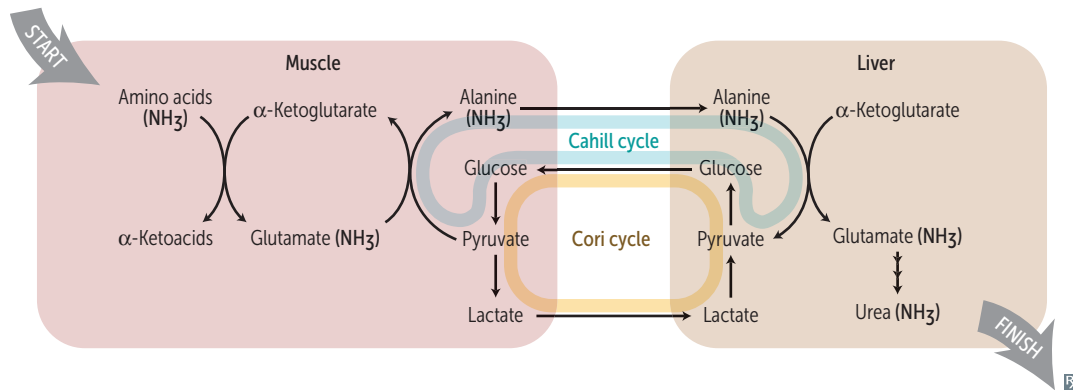
Urea cycle

Amino acid catabolism results in the formation of common metabolites (eg, pyruvate, acetyl-CoA), which serve as metabolic fuels. Excess nitrogen generated by this process is converted to urea and excreted by the kidneys.

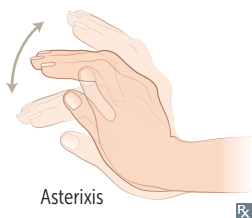
Ordinarily, Careless Crappers Are Also Frivolous About Urination.



Transport of ammonia by alanine



Hyperammonemia



Can be acquired (eg, liver disease) or hereditary (eg, urea cycle enzyme deficiencies). Results in excess NH_3 , which depletes α -ketoglutarate, leading to inhibition of TCA cycle.

Treatment: limit protein in diet.

May be given to ↓ ammonia levels:

- Lactulose to acidify the GI tract and trap NH_4^+ for excretion.
- Antibiotics (eg, rifaximin) to ↓ colonic ammoniagenic bacteria.
- Benzoate, phenylacetate, or phenylbutyrate react with glycine or glutamine, forming products that are renally excreted.

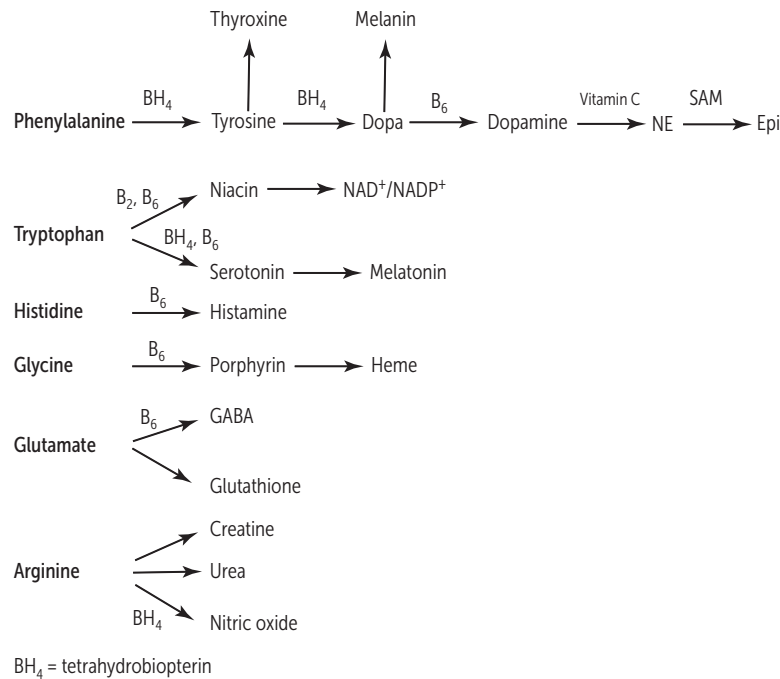
Ammonia accumulation—tremor (asterix), slurring of speech, somnolence, vomiting, cerebral edema, blurring of vision.

Ornithine transcarbamylase deficiency

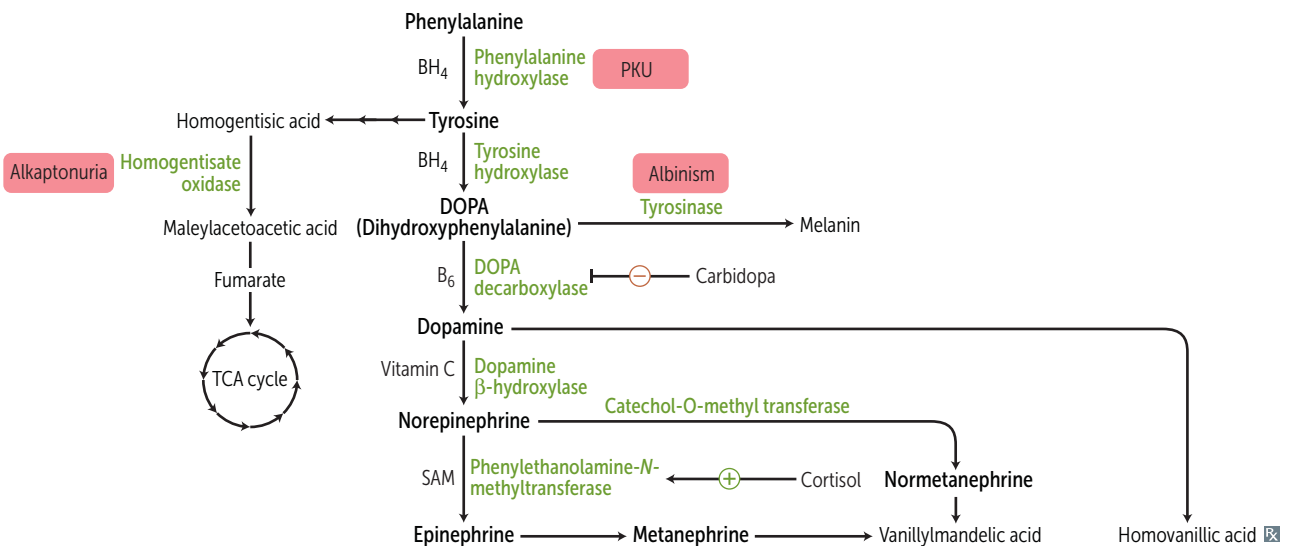
Most common urea cycle disorder. X-linked recessive (vs other urea cycle enzyme deficiencies, which are autosomal recessive). Interferes with the body's ability to eliminate ammonia. Often evident in the first few days of life, but may present later. Excess carbamoyl phosphate is converted to orotic acid (part of the pyrimidine synthesis pathway).

Findings: ↑ orotic acid in blood and urine, ↓ BUN, symptoms of hyperammonemia. No megaloblastic anemia (vs orotic aciduria).

Amino acid derivatives



Catecholamine synthesis/tyrosine catabolism



Phenylketonuria

Due to ↓ phenylalanine hydroxylase or ↓ tetrahydrobiopterin (BH₄) cofactor (malignant PKU). Tyrosine becomes essential. ↑ phenylalanine → excess phenyl ketones in urine.

Findings: intellectual disability, growth retardation, seizures, fair skin, eczema, musty body odor.

Treatment: ↓ phenylalanine and ↑ tyrosine in diet, tetrahydrobiopterin supplementation.

Maternal PKU—lack of proper dietary therapy during pregnancy. Findings in infant: microcephaly, intellectual disability, growth retardation, congenital heart defects.

Autosomal recessive. Incidence ≈ 1:10,000.

Screening occurs 2–3 days after birth (normal at birth because of maternal enzyme during fetal life).

Phenyl ketones—phenylacetate, phenyllactate, and phenylpyruvate.

Disorder of **aromatic** amino acid metabolism → musty body **odor**.

PKU patients must avoid the artificial sweetener aspartame, which contains phenylalanine.

Maple syrup urine disease

Blocked degradation of **branched** amino acids (**I**soleucine, **L**eucine, **V**aline) due to ↓ branched-chain α-ketoacid dehydrogenase (B₁). Causes ↑ α-ketoacids in the blood, especially those of leucine.

Causes severe CNS defects, intellectual disability, and death.

Treatment: restriction of isoleucine, leucine, valine in diet, and thiamine supplementation.

Autosomal recessive.

Presentation: vomiting, poor feeding, urine smells like maple syrup/burnt sugar.

I Love **V**ermont **m**aple **s**yrup from maple trees (with **B**₁**r**anches).

Alkaptonuria

Congenital deficiency of homogentisate oxidase in the degradative pathway of tyrosine to fumarate → pigment-forming homogentisic acid accumulates in tissue **A**. Autosomal recessive. Usually benign.

Findings: bluish-black connective tissue, ear cartilage, and sclerae (ochronosis); urine turns black on prolonged exposure to air. May have debilitating arthralgias (homogentisic acid toxic to cartilage).

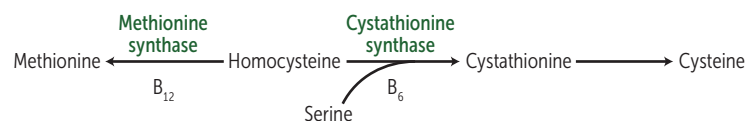
Homocystinuria

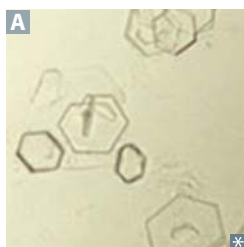
Types (all autosomal recessive):

- Cystathionine synthase deficiency (treatment: ↓ methionine, ↑ cysteine, ↑ B₆, B₁₂, and folate in diet)
- ↓ affinity of cystathionine synthase for pyridoxal phosphate (treatment: ↑↑ B₆ and ↑ cysteine in diet)
- Methionine synthase (homocysteine methyltransferase) deficiency (treatment: ↑ methionine in diet)

All forms result in excess homocysteine.

HOMOCYstinuria: ↑↑ **H**omocysteine in urine, **O**steoporosis, **M**arfanoid habitus, **O**cular changes (downward and inward lens subluxation), **C**ardiovascular effects (thrombosis and atherosclerosis → stroke and MI), **kY**phosis, intellectual disability.



Cystinuria

Hereditary defect of renal PCT and intestinal amino acid transporter that prevents reabsorption of **C**ystine, **O**rnithine, **L**ysine, and **A**rginine (**COLA**).

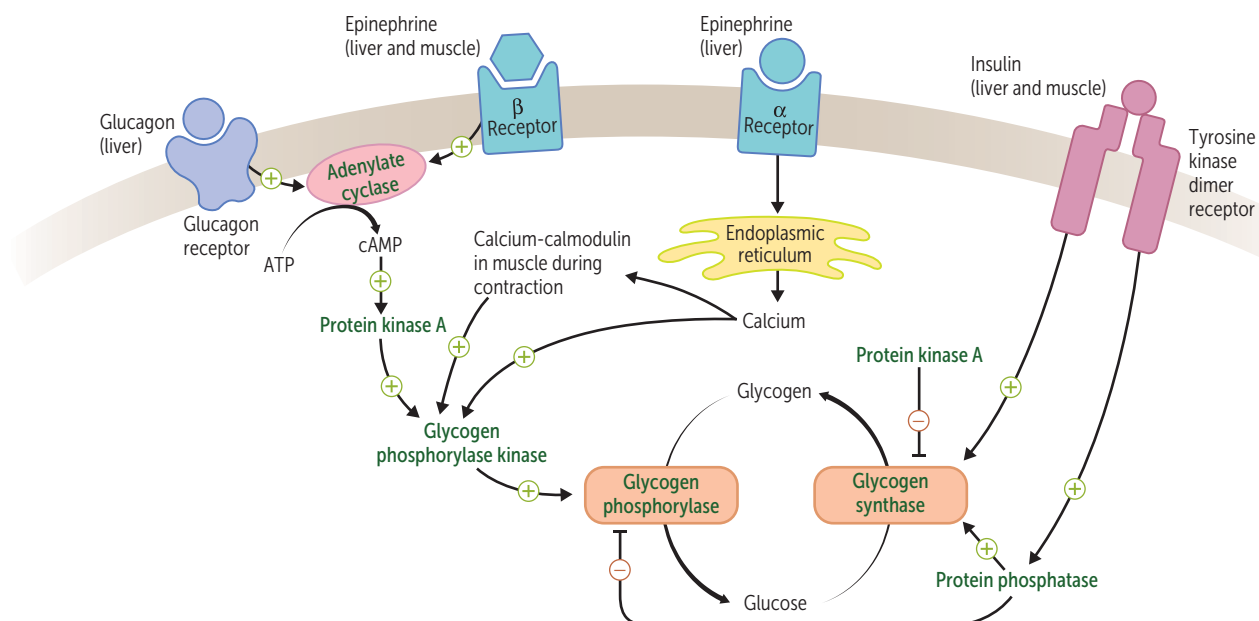
Excess cystine in the urine can lead to recurrent precipitation of hexagonal cystine stones **A**.

Treatment: urinary alkalization (eg, potassium citrate, acetazolamide) and chelating agents (eg, penicillamine) ↑ solubility of cystine stones; good hydration.

Autosomal recessive. Common (1:7000).

Urinary cyanide-nitroprusside test is diagnostic.

Cystine is made of 2 cysteines connected by a disulfide bond.

Glycogen regulation by insulin and glucagon/epinephrine

Glycogen

Branches have α -(1,6) bonds; linkages have α -(1,4) bonds.

Skeletal muscle

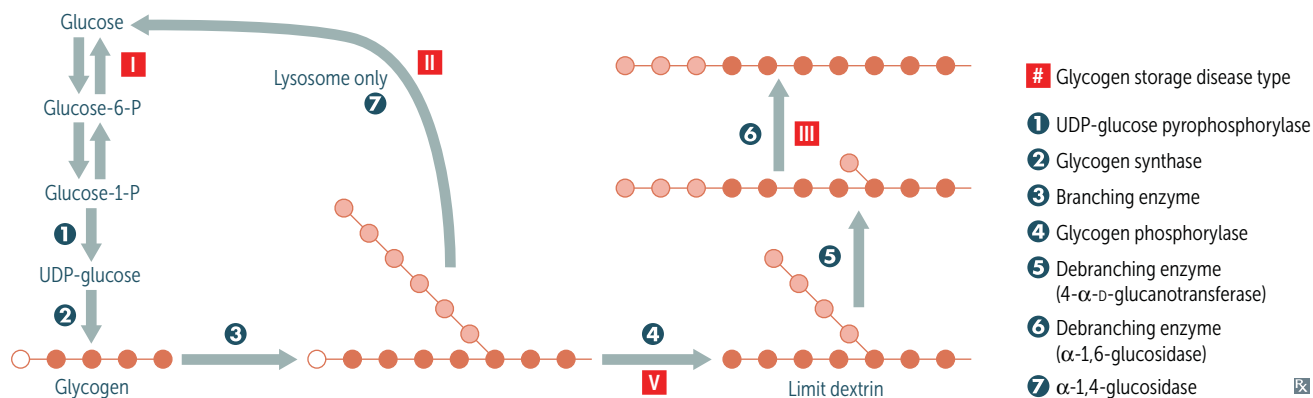
Glycogen undergoes glycogenolysis $\rightarrow$ glucose-1-phosphate $\rightarrow$ glucose-6-phosphate, which is rapidly metabolized during exercise.

Hepatocytes

Glycogen is stored and undergoes glycogenolysis to maintain blood sugar at appropriate levels.

Glycogen phosphorylase **4** liberates glucose-1-phosphate residues off branched glycogen until 4 glucose units remain on a branch. Then 4- α -D-glucanotransferase (debranching enzyme **5**) moves 3 molecules of glucose-1-phosphate from the branch to the linkage. Then α -1,6-glucosidase (debranching enzyme **6**) cleaves off the last residue, liberating glucose.

“Limit dextrin” refers to the one to four residues remaining on a branch after glycogen phosphorylase has already shortened it.



Note: A small amount of glycogen is degraded in lysosomes by **7** α -1,4-glucosidase (acid maltase).

Glycogen storage diseases

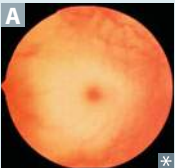
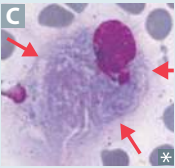
12 types, all resulting in abnormal glycogen metabolism and an accumulation of glycogen within cells. Periodic acid–Schiff stain identifies glycogen and is useful in identifying these diseases.

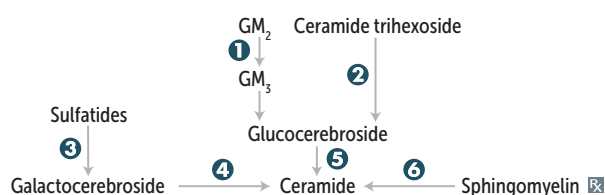
Very **P**oor **C**arbohydrate **M**etabolism.
Types I, II, III, and V are autosomal recessive.

DISEASE	FINDINGS	DEFICIENT ENZYME	COMMENTS
Von Gierke disease (type I)	Severe fasting hypoglycemia, ↑↑ G lycogen in liver, ↑ blood lactate, ↑ triglycerides, ↑ uric acid (G out), and hepatomegaly.	G lucose-6-phosphatase	Treatment: frequent oral glucose/cornstarch; avoidance of fructose and galactose Impaired gluconeogenesis and glycogenolysis
Pompe disease (type II)	Cardiomegaly, hypertrophic cardiomyopathy, hypotonia, exercise intolerance, and systemic findings lead to early death.	Lysosomal acid α-1,4 -glucosidase with α-1,6 -glucosidase activity (acid maltase)	PomPe trashes the PumP (1,4) (heart, liver, and muscle)
Cori disease (type III)	Milder form of von Gierke (type I) with normal blood lactate levels. Accumulation of limit dextrin–like structures in cytosol.	Debranching enzyme (α-1,6 -glucosidase)	Gluconeogenesis is intact
McArdle disease (type V)	↑ glycogen in muscle, but muscle cannot break it down → painful M uscle cramps, M yoglobinuria (red urine) with strenuous exercise, and arrhythmia from electrolyte abnormalities. Second-wind phenomenon noted during exercise due to ↑ muscular blood flow.	Skeletal muscle glycogen phosphorylase (M yophosphorylase)	Blood glucose levels typically unaffected McArdle = M uscle

Lysosomal storage diseases

Each is caused by a deficiency in one of the many lysosomal enzymes. Results in an accumulation of abnormal metabolic products.

DISEASE	FINDINGS	DEFICIENT ENZYME	ACCUMULATED SUBSTRATE	INHERITANCE
Sphingolipidoses				
Tay-Sachs disease 	Progressive neurodegeneration, developmental delay, “cherry-red” spot on macula A , lysosomes with onion skin, no hepatosplenomegaly (vs Niemann-Pick).	1 HeXosaminidase A (“T A y-Sa X ”)	GM ₂ ganglioside	AR
Fabry disease 	Early: Triad of episodic peripheral neuropathy, angiokeratomas B , hypohidrosis. Late: progressive renal failure, cardiovascular disease.	2 α-galactosidase A	Ceramide trihexoside	XR
Metachromatic leukodystrophy	Central and peripheral demyelination with ataxia, dementia.	3 Arylsulfatase A	Cerebroside sulfate	AR
Krabbe disease	Peripheral neuropathy, destruction of oligodendrocytes, developmental delay, optic atrophy, globoid cells.	4 Galactocerebrosidase	Galactocerebroside, psychosine	AR
Gaucher disease 	Most common. Hepatosplenomegaly, pancytopenia, osteoporosis, avascular necrosis of femur, bone crises, Gaucher cells C (lipid-laden macrophages resembling crumpled tissue paper).	5 Glucocerebrosidase (β-glucosidase); treat with recombinant glucocerebrosidase	Glucocerebroside	AR
Niemann-Pick disease 	Progressive neurodegeneration, hepatosplenomegaly, foam cells (lipid-laden macrophages) D , “cherry-red” spot on macula A .	6 Sphingomyelinase	Sphingomyelin	AR
Mucopolysaccharidoses				
Hurler syndrome	Developmental delay, gargoylism, airway obstruction, corneal clouding, hepatosplenomegaly.	α-L-iduronidase	Heparan sulfate, dermatan sulfate	AR
Hunter syndrome	Mild Hurler + aggressive behavior, no corneal clouding.	Iduronate sulfatase	Heparan sulfate, dermatan sulfate	XR



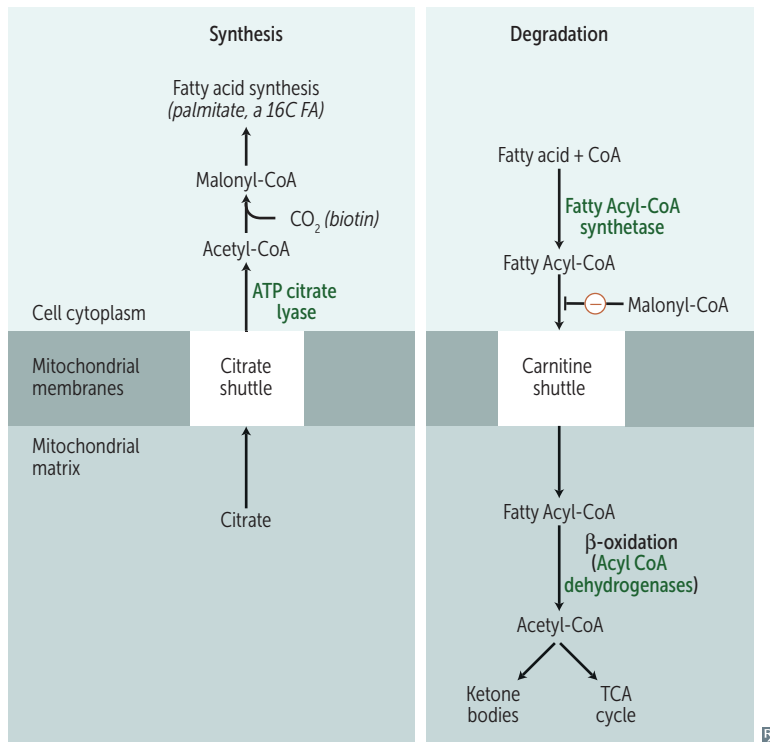
No man picks (Niemann-Pick) his nose with his **sphinger** (sphingomyelinase).

Tay-Sa**X** lacks heXosaminidase.

Hunters see clearly (no corneal clouding) and aggressively aim for the **X** (X-linked recessive).

↑ incidence of Tay-Sachs, Niemann-Pick, and some forms of Gaucher disease in Ashkenazi Jews.

Fatty acid metabolism



Fatty acid synthesis requires transport of citrate from mitochondria to cytosol. Predominantly occurs in liver, lactating mammary glands, and adipose tissue.

Long-chain fatty acid (LCFA) degradation requires carnitine-dependent transport into the mitochondrial matrix.

“**SY**trate” = **SY**nthesis.

CARnitine = **CAR**nage of fatty acids.

Systemic 1° carnitine deficiency—inherited defect in transport of LCFAs into the mitochondria → toxic accumulation. Causes weakness, hypotonia, and hypoketotic hypoglycemia.

Medium-chain acyl-CoA dehydrogenase deficiency—↓ ability to break down fatty acids into acetyl-CoA → accumulation of fatty acyl carnitines in the blood with hypoketotic hypoglycemia. Causes vomiting, lethargy, seizures, coma, liver dysfunction. Can lead to sudden death in infants or children. Treat by avoiding fasting.

Ketone bodies

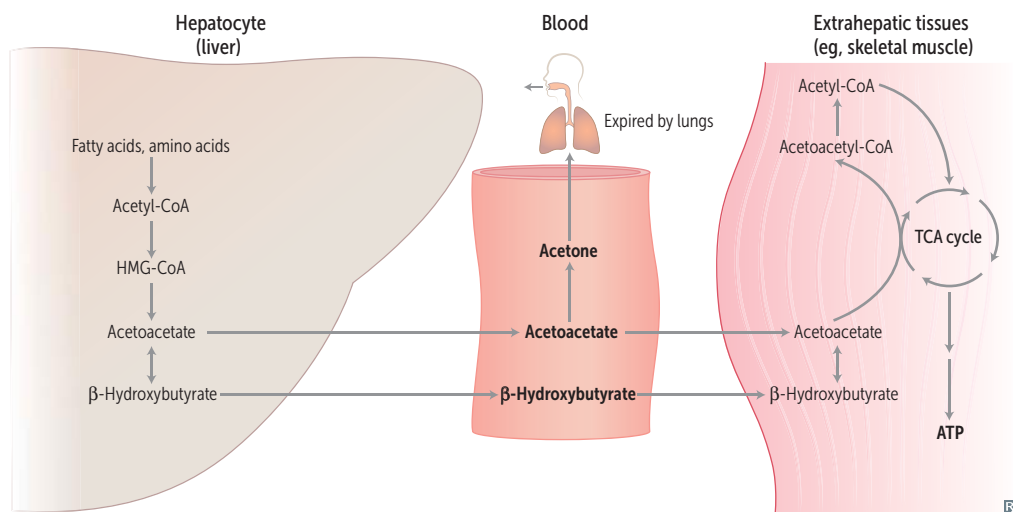
In the liver, fatty acids and amino acids are metabolized to acetoacetate and β -hydroxybutyrate (to be used in muscle and brain).

In prolonged starvation and diabetic ketoacidosis, oxaloacetate is depleted for gluconeogenesis. In alcoholism, excess NADH shunts oxaloacetate to malate. Both processes cause a buildup of acetyl-CoA, which shunts glucose and FFA toward the production of ketone bodies.

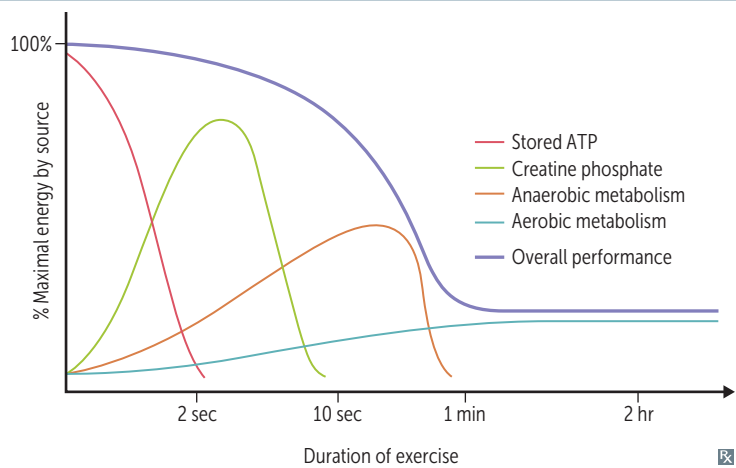
Ketone bodies: acetone, acetoacetate, β -hydroxybutyrate.

Breath smells like acetone (fruity odor).

Urine test for ketones can detect acetoacetate, but not β -hydroxybutyrate.

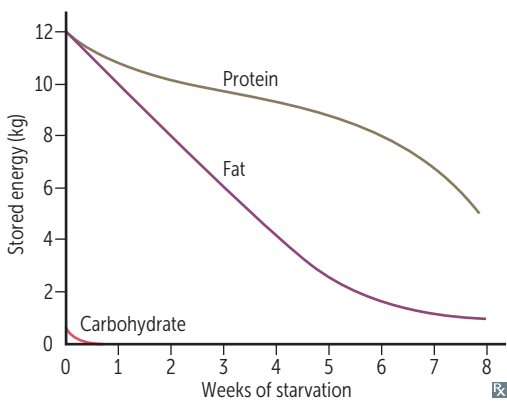


Metabolic fuel use

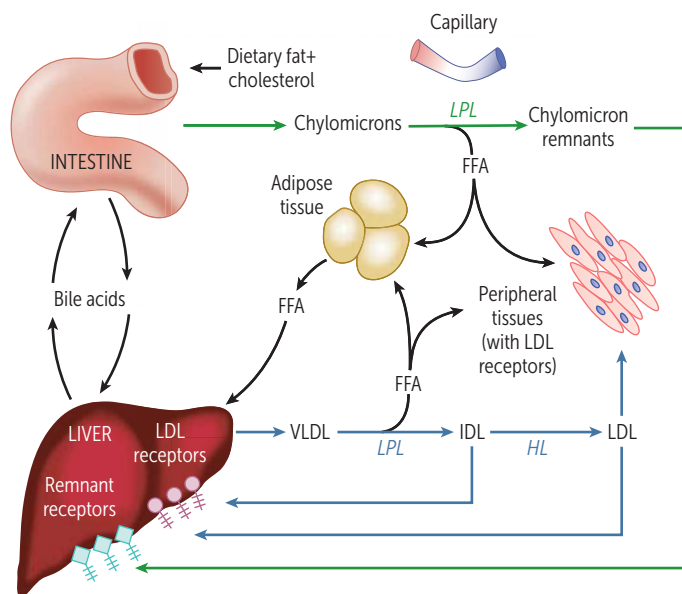


kcal (# letters = # kcal)
lg carb = 4 kcal
lg alcohol = 7 kcal
lg fatty acid = 9 kcal

Fasting and starvation	Priorities are to supply sufficient glucose to the brain and RBCs and to preserve protein.	
Fed state (after a meal)	Glycolysis and aerobic respiration.	Insulin stimulates storage of lipids, proteins, and glycogen.
Fasting (between meals)	Hepatic glycogenolysis (major); hepatic gluconeogenesis, adipose release of FFA (minor).	Glucagon and epinephrine stimulate use of fuel reserves.
Starvation days 1–3	Blood glucose levels maintained by: ▪ Hepatic glycogenolysis ▪ Adipose release of FFA ▪ Muscle and liver, which shift fuel use from glucose to FFA ▪ Hepatic gluconeogenesis from peripheral tissue lactate and alanine, and from adipose tissue glycerol and propionyl-CoA (from odd-chain FFA—the only triacylglycerol components that contribute to gluconeogenesis)	Glycogen reserves depleted after day 1. RBCs lack mitochondria and therefore cannot use ketones.
Starvation after day 3	Adipose stores (ketone bodies become the main source of energy for the brain). After these are depleted, vital protein degradation accelerates, leading to organ failure and death. Amount of excess stores determines survival time.	



Lipid transport, key enzymes

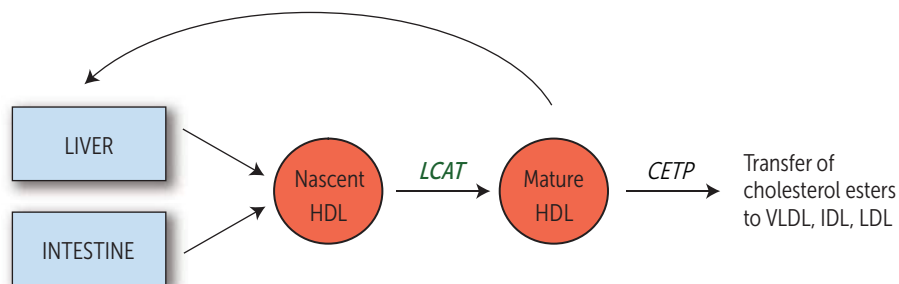


Pancreatic lipase—degradation of dietary triglycerides (TGs) in small intestine.

Lipoprotein lipase (LPL)—degradation of TGs circulating in chylomicrons and VLDLs. Found on vascular endothelial surface.

Hepatic TG lipase (HL)—degradation of TGs remaining in IDL.

Hormone-sensitive lipase—degradation of TGs stored in adipocytes.



LCAT—catalyzes esterification of $\frac{2}{3}$ of plasma cholesterol.

Cholesterol ester transfer protein (CETP)—mediates transfer of cholesterol esters to other lipoprotein particles.

Major apolipoproteins

Apolipoprotein	Function	Chylomicron	Chylomicron remnant	VLDL	IDL	LDL	HDL
E	Mediates remnant uptake (Everything E xcept LDL)	✓	✓	✓	✓		✓
A-I	Activates LCAT	✓					✓
C-II	Lipoprotein lipase C ofactor that C atalyzes C leavage	✓		✓			✓
B-48	Mediates chylomicron secretion into lymphatics	✓	✓				
B-100	Binds LDL receptor			✓	✓	✓	

Lipoprotein functions

Lipoproteins are composed of varying proportions of cholesterol, TGs, and phospholipids. LDL and HDL carry the most cholesterol.

LDL transports cholesterol from liver to tissues.

LDL is Lousy.

HDL transports cholesterol from periphery to liver.

HDL is Healthy.

Cholesterol	Needed to maintain cell membrane integrity and synthesize bile acid, steroids, and vitamin D.
Chylomicron	Delivers dietary TGs to peripheral tissues. Delivers cholesterol to liver in the form of chylomicron remnants, which are mostly depleted of their TGs. Secreted by intestinal epithelial cells.
VLDL	Delivers hepatic TGs to peripheral tissue. Secreted by liver.
IDL	Formed in the degradation of VLDL. Delivers TGs and cholesterol to liver.
LDL	Delivers hepatic cholesterol to peripheral tissues. Formed by hepatic lipase modification of IDL in the liver and peripheral tissue. Taken up by target cells via receptor-mediated endocytosis.
HDL	Mediates reverse cholesterol transport from periphery to liver. Acts as a repository for apolipoproteins C and E (which are needed for chylomicron and VLDL metabolism). Secreted from both liver and intestine. Alcohol ↑ synthesis.

Abetalipoproteinemia

Autosomal recessive. Chylomicrons, VLDL, LDL absent. Deficiency in ApoB48, ApoB100.

Affected infants present with severe fat malabsorption, steatorrhea, failure to thrive. Later manifestations include retinitis pigmentosa, spinocerebellar degeneration due to vitamin E deficiency, progressive ataxia, acanthocytosis.

Treatment: restriction of long-chain fatty acids, large doses of oral vitamin E.

Familial dyslipidemias

TYPE	INHERITANCE	PATHOGENESIS	↑ BLOOD LEVEL	CLINICAL
I—Hyper-chylomicronemia	AR	Lipoprotein lipase or apolipoprotein C-II deficiency	Chylomicrons, TG, cholesterol	Pancreatitis, hepatosplenomegaly, and eruptive/pruritic xanthomas (no ↑ risk for atherosclerosis). Creamy layer in supernatant.
II—Familial hyper-cholesterolemia	AD	Absent or defective LDL receptors	IIa: LDL, cholesterol IIb: LDL, cholesterol, VLDL	Heterozygotes (1:500) have cholesterol ≈ 300mg/dL; homozygotes (very rare) have cholesterol ≈ 700+ mg/dL. Accelerated atherosclerosis (may have MI before age 20), tendon (Achilles) xanthomas, and corneal arcus.
III—Dysbeta-lipoproteinemia	AR	Defective ApoE	Chylomicrons, VLDL	Premature atherosclerosis, tuberoeruptive xanthomas, xanthoma striatum palmare.
IV—Hyper-triglyceridemia	AD	Hepatic overproduction of VLDL	VLDL, TG	Hypertriglyceridemia (> 1000 mg/dL) can cause acute pancreatitis.